

COURSE NOTES

Lie Theory

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March 8, 2026

Contents

1	Lecture 1 Where Lie algebras come from	2
1.1	Introduction	2
1.2	Where do Lie algebras come from?	5
1.2.1	They come from associative algebras	5
1.2.2	They come from involutions of central simple algebras	6
1.2.3	They come from the derivation of any algebra	12
1.2.4	They come from brackets	18
1.2.5	They come from Lie groups	20
1.3	Basic concepts of Lie algebra	22
2	Lecture 2 Engel Theorem	25
2.1	Engel Theorem the first type	25
2.2	Engel Theorem the second type	26
2.3	Engel Theorem the third type	30
3	Lecture 3: Lie's theorem and Jordan-Chevalley decomposition	36
3.1	Lie's theorem	36
3.2	Generalized eigenvectors and generalized eigenspaces	42
4	Homework	44
4.1	Homework week 1	44
4.2	Homework week 2	50

Chapter 1

Lecture 1 Where Lie algebras come from

1.1 Introduction

Lie algebras were originally introduced by Sophus Lie in the 1870s to study the concept of continuous transformation groups. Lie's motivation was to develop a theory for differential equations analogous to Galois theory for algebraic equations. He discovered that the infinitesimal transformations of a continuous group form a linear space equipped with a bracket operation, now known as a Lie algebra. This linearization allows complex global geometric problems to be translated into algebraic problems, making them more tractable.

Let F be a field (mostly $\text{char } F = 0$, even $F = \mathbb{C}$).

Definition 1.1.1. A linear algebra A is a vector space over F , with a binary bilinear operation:

- *binary operation:* $A \times A \rightarrow A$, $(a, b) \mapsto ab$.
- *bilinear:* this operation is linear in each variable. In other words, $\forall \alpha \in F$ and $a, a', a'', b, b', b'' \in A$, we have:

$$(\alpha a)b = a(\alpha b) = \alpha(ab)$$

$$(a' + a'')b = a'b + a''b$$

$$a(b' + b'') = ab' + ab''$$

The algebra A is associative if $\forall a, b, c \in A$, $(ab)c = a(bc)$.

Example 1.1.2. $M_n(F)$, the algebra of $n \times n$ matrices over F .

Example 1.1.3. Let V be a vector space, $\text{End}_F(V) = \{\text{linear transformations } \varphi : V \rightarrow V\}$ is an associative algebra with the composition operation $(\phi\psi)(v) = \phi(\psi(v))$.

If $\dim V < \infty$, when we fix a basis, then we get a matrix representation of φ in our fix basis.

The most important thing everyone should know is Wedderburn's theorem.

Theorem 1.1.4 (Wedderburn-Artin). *Let A be a semisimple Artinian ring (with 1). Then A is a direct sum of matrix rings over division rings. Specifically, if ${}_A A \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$ where $S_1, \dots, S_r$ are non-isomorphic simple modules occurring with multiplicities $n_1, \dots, n_r$, then*

$$A \simeq M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

where $D_i = \text{End}_A(S_i)^{op}$. This is a ring isomorphism, and each $M_{n_i}(D_i)$ is a 2-sided ideal.

Proof. First, since A is a semisimple ring, its left regular module ${}_A A$ can be decomposed into a direct sum of simple left A -modules:

$${}_A A \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$$

where $S_1, \dots, S_r$ are pairwise non-isomorphic simple modules.

By Schur's lemma, the endomorphism ring of a simple module is a division ring, so $D_i = \text{End}_A(S_i)^{op}$ is a division ring.

Next, we compute the endomorphism ring of the left regular module $\text{End}_A({}_A A)$. On one hand, for any ring A with 1, we have the ring isomorphism $\text{End}_A({}_A A) \simeq A^{op}$. On the other hand, using our direct sum decomposition:

$$\text{End}_A({}_A A) \simeq \text{End}_A(S_1^{n_1}) \oplus \cdots \oplus \text{End}_A(S_r^{n_r})$$

This decomposition holds because $\text{Hom}_A(S_i^{n_i}, S_j^{n_j}) = 0$ whenever $i \neq j$.

For each component in the direct sum, we can rewrite the endomorphism ring using matrices over D_i :

$$\text{End}_A(S_i^{n_i}) \simeq M_{n_i}(D_i)^{op} \simeq M_{n_i}(D_i^{op})$$

Substituting this back into our equation for A^{op} , we obtain:

$$A^{op} \simeq M_{n_1}(D_1^{op}) \oplus \cdots \oplus M_{n_r}(D_r^{op})$$

To recover the ring A , we take the opposite ring on both sides. Recall the property that $M_n(R)^{op} \simeq M_n(R^{op})$. Applying this to each block yields:

$$A \simeq (A^{op})^{op} \simeq \left(\bigoplus_{i=1}^r M_{n_i}(D_i^{op}) \right)^{op} \simeq \bigoplus_{i=1}^r M_{n_i}(D_i)$$

Thus, the isomorphism is established. $\square$

Remark 1.1.5. If the base field F is algebraically closed, every finite-dimensional division algebra over F is equal to F . Hence the theorem simplifies to

$$A \cong \prod_{i=1}^r M_{n_i}(F).$$

Example 1.1.6. Let G be a finite group and let $\mathbb{C}[G]$ be its group algebra. By Maschke's theorem $\mathbb{C}[G]$ is semisimple. Therefore

$$\mathbb{C}[G] \cong \prod_{i=1}^r M_{n_i}(\mathbb{C}),$$

where n_i are the dimensions of the irreducible complex representations of G .

Definition 1.1.7. Suppose F is algebraically closed. An algebra A is simple if

1. $\nexists I \triangleleft A$ and $I \neq 0$.
2. $A \cdot A \neq (0)$.

Corollary 1.1.8. Any simple finite dimensional associative algebra is isomorphic to $M_n(F)$ for some n .

Definition 1.1.9 (Jacobson Radical). Let A be an associative algebra with identity over a field F . The Jacobson radical of A , denoted by $\text{Rad}(A)$ or $J(A)$, is defined as the intersection of all maximal left ideals of A :

$$\text{Rad}(A) = \bigcap_{M \text{ maximal left ideal of } A} M.$$

Proposition 1.1.10. Let A be an associative algebra with identity. Then

$$\text{Rad}(A) = \bigcap_{S \text{ simple left } A\text{-module}} \text{Ann}_A(S),$$

where

$$\text{Ann}_A(S) = \{a \in A \mid aS = 0\}$$

is the annihilator of S in A .

Proposition 1.1.11 (Equivalent Characterization). Let A be an associative algebra with identity. Then an element $a \in A$ lies in $\text{Rad}(A)$ if and only if for every $x \in A$ the element

$$1 - ax$$

is invertible in A .

Proposition 1.1.12. *The Jacobson radical $\text{Rad}(A)$ is a two-sided ideal of A .*

Definition 1.1.13 (Semisimple Algebra). *An associative algebra A is called semisimple if*

$$\text{Rad}(A) = 0.$$

1.2 Where do Lie algebras come from?

Definition 1.2.1. *A Lie algebra is a vector space L over a field F , with a binary operation $[a, b] \in L$, such that*

1. $[a, b]$ is bilinear.
2. $[a, b] = -[b, a]$.
3. $[a, [b, c]] + [c, [a, b]] + [b, [c, a]] = 0$, this is called Jacobi identity.

1.2.1 They come from associative algebras

Let A be an associative algebra. $\forall a, b \in A$, define $[a, b] = ab - ba$. Check that $(A, [,])$ is a Lie algebra:

Proof. Bilinearity and skew-symmetry are clear. For Jacobi identity:

$$\begin{aligned} & [a, [b, c]] + [b, [c, a]] + [c, [a, b]] \\ &= a(bc - cb) - (bc - cb)a + b(ca - ac) - (ca - ac)b + c(ab - ba) - (ab - ba)c \\ &= abc - acb - bca + cba + bca - bac - cab + acb + cab - cba - abc + bac \\ &= 0. \end{aligned}$$

□

We denote this new algebra $(A, [,])$ by $A^{(-)}$.

Suppose that $L \subseteq A$ and $[L, L] \subseteq L$. Then $L \leq A^{(-)}$.

Example 1.2.2. $M_n(F)^{(-)} = \mathfrak{gl}(n)$. We call it the general linear Lie algebra. $[L, L] = \{\sum_i [a_i, b_i]\} \subseteq L$. Clearly $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq \mathfrak{gl}(n)$ and for $n \geq 2$, $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq 0$.

Example 1.2.3. Focus on $[\mathfrak{gl}(n), \mathfrak{gl}(n)]$, all matrices are of trace 0. This is called the special linear Lie algebra, denoted by $\mathfrak{sl}(n)$.

Example 1.2.4. *If the base field changed, some properties of the Lie algebra may change. For example, if $\text{char } F = 2$, then $[a, b] = [b, a]$. The Lie bracket is no longer skew-symmetric but actually symmetric.*

Consider $[\mathfrak{gl}(n), \mathfrak{gl}(n)] = \mathfrak{sl}(n)$. If $\text{char } F = 0$ or $\text{char } F \nmid n$, then $\mathfrak{sl}(n)$ is a simple Lie algebra. If $\text{char } F \mid n$, then $\mathfrak{sl}(n)$ is not simple.

Definition 1.2.5. Z is the center of Lie algebra L if $Z = \{z \in L \mid [z, L] = 0\}$.

1.2.2 They come from involutions of central simple algebras

We can also find Lie algebras from the involution of a central simple algebra. Let A be central simple algebra and $*$: $A \rightarrow A$ be a linear transformation. $*$ is called a an involution if $(a^*)^* = a$ and $(ab)^* = b^*a^*$, $\forall a, b \in A$.

Proposition 1.2.6. *Let A be a central simple algebra over a field F equipped with an involution $*$: $A \rightarrow A$. Then the set*

$$L = \{x \in A \mid *(x) = -x\}$$

forms a Lie algebra under the commutator bracket

$$[x, y] = xy - yx.$$

Proof. First observe that any associative algebra A becomes a Lie algebra with the commutator bracket

$$[x, y] = xy - yx.$$

Let

$$L = \{x \in A \mid *(x) = -x\}.$$

We show that L is closed under the Lie bracket.

Take $x, y \in L$. Then

$$*(x) = -x, \quad *(y) = -y.$$

Since $*$ is an involution, it satisfies

$$*(xy) = *(y) * (x).$$

Hence

$$*([x, y]) = *(xy - yx) = *(y) * (x) - *(x) * (y).$$

Substituting $*(x) = -x$ and $*(y) = -y$, we obtain

$$*([x, y]) = (-y)(-x) - (-x)(-y) = yx - xy = -[x, y].$$

Thus $[x, y] \in L$. Therefore L is closed under the commutator bracket.

Since the commutator bracket in any associative algebra satisfies bilinearity, antisymmetry, and the Jacobi identity, it follows that L is a Lie algebra. $\square$

This construction explains why many classical Lie algebras arise from central simple algebras with involution. In particular, the classical Lie algebras of Cartan–Killing type can be obtained in this way.

Example 1.2.7 (Orthogonal involution). *Let*

$$A = M_n(F)$$

and define the involution

$$*(X) = X^T.$$

Then

$$L = \{X \in M_n(F) \mid X^T = -X\}.$$

This is the Lie algebra of skew-symmetric matrices, which is the orthogonal Lie algebra

$$\mathfrak{so}_n(F).$$

Example 1.2.8 (Symplectic involution). *Let*

$$A = M_{2n}(F)$$

and let

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Define

$$*(X) = J^{-1}X^TJ.$$

Then

$$L = \{X \in M_{2n}(F) \mid *(X) = -X\}$$

is the symplectic Lie algebra

$$\mathfrak{sp}_{2n}(F).$$

Example 1.2.9 (Unitary involution). *Let K/F be a quadratic field extension with Galois involution $a \mapsto \bar{a}$. Consider*

$$A = M_n(K)$$

with involution

$$*(X) = \overline{X}^T.$$

Then

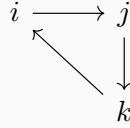
$$L = \{X \in M_n(K) \mid *(X) = -X\}$$

is the unitary Lie algebra

$$\mathfrak{su}_n.$$

Therefore classical Lie algebras arise from pairs $(A, *)$ where A is a central simple algebra and $*$ is an involution. This provides an algebraic framework for the Cartan–Killing classification of classical Lie algebras.

Example 1.2.10. The Quaternion algebra is an associative algebra with the involution $a \mapsto a^*$. It is 4 dimensional central simple algebra with standard basis $1, i, j, k$, where the relations are $i^2 = j^2 = k^2 = -1, ij = k, jk = i, ki = j$ as the following graph:



We use $\mathbb{H}$ to denote the Quaternion algebra when we choose $\mathbb{R}$ as the base field. i.e. $\mathbb{H} = \{a_0 1 + a_1 i + a_2 j + a_3 k \mid a_0, a_1, a_2, a_3 \in \mathbb{R}\}$. $\forall x \in \mathbb{H}$ where $x = a_0 1 + a_1 i + a_2 j + a_3 k$, $*$ is defined as $x^* = a_0 1 - a_1 i - a_2 j - a_3 k$ and $x^{-1} = \frac{1}{a_0^2 + a_1^2 + a_2^2 + a_3^2} (a_0 1 - a_1 i - a_2 j - a_3 k)$. Hence $\mathbb{H}$ is a division algebra.

If the base field is algebraically closed, $F = \overline{F}$. We can denote the Quaternion algebra by 2×2 matrices

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix}$$

Where $\alpha, \beta, \gamma, \xi \in F$. In this case it is no longer a division algebra ($a_0^2 + a_1^2 + a_2^2 + a_3^2$ may equal to 0), but just $M_2(F)$. And in this case the involution $*$ is defined as:

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix} \mapsto \begin{pmatrix} \xi & -\beta \\ -\gamma & \alpha \end{pmatrix}$$

Which is the Symplectic involution.

Example 1.2.11. From what we discussed. For a $2n \times 2n$ matrix $A = \begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix}$ in which A_1, A_2, A_3, A_4 are $n \times n$ matrices, the unitary involution is defined as:

$$A^* = \begin{pmatrix} A_4^t & -A_3^t \\ -A_2^t & A_1^t \end{pmatrix}$$

And the unitary involution is defined as:

$$A^* = \overline{A}^t$$

When $n = 1$ we have two different type, and we will get two different Lie algebras. But they are isomorphic.

Proposition 1.2.12. *Let F be a field of characteristic $\neq 2$, and let*

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in M_2(F).$$

Define an involution $$ on $M_2(F)$ by*

$$X^* = J^{-1}X^tJ.$$

Then the Lie algebra of $$ -skew elements,*

$$K(M_2(F), *) = \{X \in M_2(F) \mid X^* = -X\},$$

is exactly $\mathfrak{sl}_2(F)$.

Proof. Let

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(F).$$

Since

$$J^{-1} = -J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

we compute

$$X^* = J^{-1}X^tJ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Now the condition $X^* = -X$ becomes

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix}.$$

Hence $d = -a$, while b and c are arbitrary. Therefore

$$K(M_2(F), *) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \mid a, b, c \in F \right\}.$$

But this is precisely the set of all 2×2 matrices of trace 0, namely

$$\mathfrak{sl}_2(F) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \mid a, b, c \in F \right\}.$$

Thus

$$K(M_2(F), *) = \mathfrak{sl}_2(F).$$

Since $K(M_2(F), *)$ is closed under the commutator bracket, it follows that the Lie algebra obtained from the symplectic involution on $M_2(F)$ is $\mathfrak{sl}_2(F)$. $\square$

Example 1.2.13. Let K/F be a quadratic field extension, and let $a \mapsto \bar{a}$ denote the nontrivial F -automorphism of K . Consider the algebra $M_2(K)$ and define an involution $*$ by

$$X^* = \overline{X}^t.$$

Then $*$ is a unitary involution on $M_2(K)$.

The Lie algebra of $*$ -skew elements is

$$K(M_2(K), *) = \{X \in M_2(K) \mid X^* = -X\}.$$

If

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then

$$X^* = \overline{X}^t = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}.$$

Thus the condition $X^* = -X$ is equivalent to

$$\bar{a} = -a, \quad \bar{d} = -d, \quad \bar{c} = -b.$$

Hence every element of $K(M_2(K), *)$ has the form

$$X = \begin{pmatrix} a & b \\ -\bar{b} & d \end{pmatrix}, \quad \text{with } \bar{a} = -a, \bar{d} = -d.$$

This is the unitary Lie algebra associated with the standard hermitian form.

Its derived Lie algebra consists of those elements with trace 0, namely

$$\mathfrak{su}_2(K/F) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & -a \end{pmatrix} \mid a, b \in K, \bar{a} = -a \right\}.$$

After extending scalars to a splitting field (for instance to an algebraic closure of F), this Lie algebra becomes isomorphic to $\mathfrak{sl}_2$. Therefore the Lie algebra arising from a unitary involution in the 2×2 case is of type A_1 .

Remark 1.2.14. Why does this Lie algebra become $\mathfrak{sl}_2$? The reason is that unitary Lie algebras are forms of special linear Lie algebras. In the present 2×2 case, the corresponding type is A_1 . Over a field where the quadratic extension splits, the involution becomes equivalent to the ordinary transpose situation, and the resulting special unitary Lie algebra becomes the split simple Lie algebra of type A_1 , namely

$$\mathfrak{sl}_2.$$

Thus, although over the ground field one naturally writes the Lie algebra as $\mathfrak{su}_2$, its split form is precisely $\mathfrak{sl}_2$.

Example 1.2.15 (The unitary involution on $M_2(\mathbb{C})$ and the Lie algebra $\mathfrak{su}_2$). Let $F = \mathbb{R}$ and $K = \mathbb{C}$, and let $\bar{}$ denote complex conjugation. Consider the algebra $M_2(\mathbb{C})$ equipped with the involution

$$X^* = \overline{X}^t.$$

This is the standard unitary involution on $M_2(\mathbb{C})$ relative to the extension $\mathbb{C}/\mathbb{R}$. We first determine the Lie algebra of $*$ -skew elements:

$$K(M_2(\mathbb{C}), *) = \{X \in M_2(\mathbb{C}) \mid X^* = -X\}.$$

Let

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{C}).$$

Then

$$X^* = \overline{X}^t = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}.$$

Thus the condition $X^* = -X$ is equivalent to

$$\bar{a} = -a, \quad \bar{d} = -d, \quad \bar{c} = -b.$$

Hence a and d are purely imaginary, and $c = -\bar{b}$. Therefore

$$K(M_2(\mathbb{C}), *) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & d \end{pmatrix} \mid a, d \in i\mathbb{R}, b \in \mathbb{C} \right\}.$$

This is the unitary Lie algebra $\mathfrak{u}_2$.

Its derived Lie algebra is the trace-zero part, namely

$$\mathfrak{su}_2 = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & -a \end{pmatrix} \mid a \in i\mathbb{R}, b \in \mathbb{C} \right\}.$$

Indeed, $\text{tr}([X, Y]) = 0$ for all $X, Y \in M_2(\mathbb{C})$, so every commutator in $\mathfrak{u}_2$ has trace 0. Conversely, in this 2×2 case, the trace-zero subspace is generated by commutators, and hence it is exactly the derived Lie algebra.

To see that $\mathfrak{su}_2$ is 3-dimensional over $\mathbb{R}$, write

$$a = it, \quad b = u + iv, \quad t, u, v \in \mathbb{R}.$$

Then every element of $\mathfrak{su}_2$ has the form

$$\begin{pmatrix} it & u + iv \\ -u + iv & -it \end{pmatrix}, \quad t, u, v \in \mathbb{R}.$$

Therefore

$$\dim_{\mathbb{R}} \mathfrak{su}_2 = 3.$$

A convenient real basis is

$$H = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

One checks directly that

$$[H, X] = 2Y, \quad [H, Y] = -2X, \quad [X, Y] = 2H,$$

so $\mathfrak{su}_2$ is a 3-dimensional real Lie algebra.

Now we explain why this Lie algebra becomes $\mathfrak{sl}_2(\mathbb{C})$ after extending scalars from $\mathbb{R}$ to $\mathbb{C}$. Consider the complexification

$$\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C}.$$

Since H, X, Y form a real basis of $\mathfrak{su}_2$, they also form a complex basis of $\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C}$. Define

$$h = -iH = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad e = \frac{1}{2}(X - iY) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \frac{1}{2}(-X - iY) = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}.$$

These elements lie in $\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C}$ and satisfy

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

Thus they form a standard $\mathfrak{sl}_2$ -triple, and therefore

$$\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{sl}_2(\mathbb{C}).$$

In other words, $\mathfrak{su}_2$ is a real form of $\mathfrak{sl}_2(\mathbb{C})$. Hence the Lie algebra arising from the unitary involution on $M_2(\mathbb{C})$ is of type A_1 .

Remark 1.2.16. This explains why, in the 2×2 case, the Lie algebra arising from a symplectic involution is isomorphic to the one of type A_1 . Equivalently, in this lowest-rank case one has the accidental isomorphism

$$C_1 \cong A_1.$$

1.2.3 They come from the derivation of any algebra

Let A be any algebra.

Definition 1.2.17. A derivation of A is a linear map $d : A \rightarrow A$ such that $d(ab) = d(a)b + ad(b)$ for all $a, b \in A$.

Example 1.2.18. Let $A = F[t_1, \dots, t_n]$ be the polynomial algebra. We have the formal derivative ∂ where $\partial(t_i^n) = nt_i^{n-1}$ and $\partial(fg) = f\partial(g) + \partial(f)g$. Let $d = \sum_{i=1}^n f_i(t_1, \dots, t_n) \frac{\partial}{\partial t_i}$, this is a derivation of A .

$$d(f) = \sum_{i=1}^n f_i(t_1, \dots, t_n) \frac{\partial f}{\partial t_i}$$

If d_1, d_2 are derivations of A , then $[d_1, d_2]$ is again a derivation of A . We can check it by the following calculation:

$$\begin{aligned} [d_1, d_2](ab) &= d_1(d_2(ab)) - d_2(d_1(ab)) \\ &= d_1(d_2(a)b + ad_2(b)) - d_2(d_1(a)b + ad_1(b)) \\ &= d_1(d_2(a))b + d_1(ad_2(b)) + d_2(d_1(a))b + d_2(ad_1(b)) \\ &= d_1(d_2(a))b + d_2(d_1(a))b + d_1(ad_2(b)) + d_2(ad_1(b)) \\ &= d_1(d_2(a))b + d_2(d_1(a))b + a[d_1, d_2](b) \\ &= [d_1, d_2](a)b + a[d_1, d_2](b) \\ &= [d_1, d_2](ab) \end{aligned}$$

Therefore $[d_1, d_2]$ is a derivation of A .

The above example means that:

$$\text{Der}(A) = \{ d : A \rightarrow A \mid d \text{ is a derivation} \} < \text{End}_F(A)^{(-)}$$

In particular, when $A = F[t_1, \dots, t_n]$, we have:

$$\text{Der}(A) = \left\{ \sum_{i=1}^n f_i(t_1, \dots, t_n) \frac{\partial}{\partial t_i} \mid f_i(t_1, \dots, t_n) \in A \right\} < \text{End}_F(A)^{(-)}$$

We denote this algebra by $W(n)$. It is called the n -th Witt algebra. It is a simple, infinite dimensional Lie algebra (when $\text{char } F = 0$).

Thus every element of $W(n)$ has the form

$$D = \sum_{i=1}^n f_i \partial_i, \quad f_i \in A.$$

Definition 1.2.19. *The divergence map is defined by*

$$\operatorname{div} : W(n) \rightarrow A, \quad \operatorname{div} \left(\sum_{i=1}^n f_i \partial_i \right) = \sum_{i=1}^n \partial_i(f_i).$$

Its kernel is the special Lie algebra

$$S(n) := \ker(\operatorname{div}) = \left\{ \sum_{i=1}^n f_i \partial_i \in W(n) \mid \sum_{i=1}^n \partial_i(f_i) = 0 \right\}.$$

Equivalently, $S(n)$ is the Lie algebra of polynomial vector fields preserving the standard volume form

$$\omega = dx_1 \wedge \cdots \wedge dx_n.$$

A basic family of elements of $S(n)$ is given by

$$D_{ij}(f) := \partial_j(f) \partial_i - \partial_i(f) \partial_j, \quad 1 \leq i < j \leq n, \quad f \in A,$$

since

$$\operatorname{div}(D_{ij}(f)) = \partial_i \partial_j(f) - \partial_j \partial_i(f) = 0.$$

One can check

$$S(n) = \langle D_{ij}(f) \mid 1 \leq i < j \leq n, f \in A \rangle_{\operatorname{Lie}}$$

These elements generate $S(n)$ as a Lie algebra.

Definition 1.2.20. *The grading of $W(n)$ is defined by Give A its standard grading by declaring $\deg x_i = 1$. Then $W(n)$ becomes a $\mathbb{Z}$ -graded Lie algebra with*

$$\deg(x^a \partial_i) = |a| - 1, \quad |a| := a_1 + \cdots + a_n,$$

so that

$$W(n) = \bigoplus_{k \geq -1} W(n)_k.$$

and

$$W(n)_0 = \operatorname{span}_F \{x_j \partial_i \mid 1 \leq i, j \leq n\} \cong \mathfrak{gl}_n(F).$$

Since the divergence map is homogeneous of degree 0, the subalgebra $S(n)$ is graded:

$$S(n) = \bigoplus_{k \geq -1} S(n)_k, \quad S(n)_k = S(n) \cap W(n)_k.$$

Moreover,

$$S(n)_{-1} = W(n)_{-1},$$

and

$$S(n)_0 = \left\{ \sum_{i,j} a_{ij} x_j \partial_i \mid \sum_i a_{ii} = 0 \right\} \cong \mathfrak{sl}_n(F),$$

because

$$\operatorname{div} \left(\sum_{i,j} a_{ij} x_j \partial_i \right) = \sum_i a_{ii}.$$

In characteristic 0, the special algebra is perfect:

$$[S(n), S(n)] = S(n).$$

Hence

$$S(n)^{(1)} = S(n).$$

For $n \geq 2$, the Lie algebra $S(n)$ is simple. Therefore, in characteristic 0 one has

$$W(n) \supset S(n) = S(n)^{(1)},$$

where both algebras are infinite-dimensional, $W(n)$ is the full Lie algebra of polynomial vector fields, and $S(n)$ is its simple divergence-zero subalgebra.

Remark 1.2.21. *In characteristic 0, the chain $W(n) \supset S(n) \supset S(n)^{(1)}$ is thus less informative than in positive characteristic, since the last inclusion is an equality.*

Let A be an associative algebra. Fix an element $a \in A$, we define : $\operatorname{ad}(a) : A \rightarrow A : b \mapsto [a, b]$. This is a derivation of A .

Definition 1.2.22. *Let L be a Lie algebra. The adjoint map is defined as:*

$$\operatorname{ad} : L \rightarrow \operatorname{End}_F(L)^{(-)}, a \mapsto \operatorname{ad}(a)$$

where $\operatorname{ad}(a) : L \rightarrow L : b \mapsto [a, b]$ is an inner derivation of L . We also call $\operatorname{ad}(a)$ an adjoint operator.

Note that $\operatorname{ad}(a) \in \operatorname{Der}(L)$ check it by the following calculation:

$$\begin{aligned} \operatorname{ad}(a)([b, c]) &= \operatorname{ad}(a)(bc - cb) \\ &= [a, bc] - [a, cb] \\ &= [a, b]c + b[a, c] - [a, c]b - c[a, b] \\ &= [\operatorname{ad}(a)(b), c] + [b, \operatorname{ad}(a)(c)] \end{aligned}$$

”Inner” here comes from that $\operatorname{ad}(L) \triangleleft \operatorname{Der}(L)$ since $\forall d \in \operatorname{Der}(L)$, we have $[d, \operatorname{ad}(a)] =$

$\text{ad}(\text{ad}(a))$. Check it by the following calculation:

$$\begin{aligned} [d, \text{ad}(a)](x) &= d([a, x]) - [a, d(x)] \\ &= [d(a), x] + [a, d(x)] - [a, d(x)] \\ &= [d(a), x] \\ &= \text{ad}(d(a))(x). \end{aligned}$$

Remark 1.2.23 (Exponentials of nilpotent inner derivations). *Let L be a Lie algebra over a field F , and let*

$$d \in \text{ad}(L) \subseteq \text{Der}(L).$$

Thus there exists $a \in L$ such that

$$d = \text{ad}(a), \quad d(x) = [a, x] \quad \text{for all } x \in L.$$

Assume that d is nilpotent, so that $d^N = 0$ for some integer $N \geq 1$. Then one defines the exponential of d by

$$\exp(d) := \sum_{n=0}^{\infty} \frac{d^n}{n!}.$$

Since d is nilpotent, this is actually a finite sum:

$$\exp(d) = I + d + \frac{d^2}{2!} + \cdots + \frac{d^{N-1}}{(N-1)!}.$$

Hence $\exp(d)$ is a well-defined linear endomorphism of L .

Moreover, $\exp(d)$ is invertible, with inverse

$$\exp(d)^{-1} = \exp(-d).$$

Since d is a derivation, $\exp(d)$ is in fact a Lie algebra automorphism:

$$\exp(d)([x, y]) = [\exp(d)(x), \exp(d)(y)] \quad \text{for all } x, y \in L.$$

Therefore

$$\exp(d) \in \text{Aut}(L).$$

In particular, if $d = \text{ad}(a)$, then

$$\exp(\text{ad}(a))(x) = x + [a, x] + \frac{1}{2!}[a, [a, x]] + \frac{1}{3!}[a, [a, [a, x]]] + \cdots.$$

When $\text{ad}(a)$ is nilpotent, this sum is finite. The automorphism

$$\exp(\text{ad}(a))$$

is called an inner automorphism arising from the nilpotent inner derivation $\text{ad}(a)$.

A fundamental example is the matrix Lie algebra $L = \mathfrak{gl}_n(F)$. For $A \in \mathfrak{gl}_n(F)$, one has

$$\mathrm{ad}(A)(X) = [A, X] = AX - XA,$$

and

$$\exp(\mathrm{ad}(A))(X) = e^A X e^{-A}.$$

Thus, in the matrix case, the exponential of an inner derivation is exactly conjugation by the matrix exponential.

Proposition 1.2.24. *Let L be a Lie algebra over a field F , and let $d \in \mathrm{Der}(L)$ be nilpotent. Then*

$$\exp(d) := \sum_{n=0}^{\infty} \frac{d^n}{n!}$$

is a Lie algebra automorphism of L .

Proof. Since d is nilpotent, there exists $N \geq 1$ such that $d^N = 0$. Hence

$$\exp(d) = I + d + \frac{d^2}{2!} + \cdots + \frac{d^{N-1}}{(N-1)!}$$

is a finite sum, so it is a well-defined linear endomorphism of L .

Moreover,

$$\exp(d) \exp(-d) = \exp(-d) \exp(d) = I,$$

because d commutes with itself and the usual exponential identity holds for polynomials in d . Thus $\exp(d)$ is invertible, with inverse $\exp(-d)$.

It remains to show that $\exp(d)$ preserves the Lie bracket. Define

$$\phi_t := \exp(td) = \sum_{n=0}^{\infty} \frac{t^n d^n}{n!},$$

where the sum is again finite since d is nilpotent. For $x, y \in L$, set

$$F(t) := \phi_t([x, y]), \quad G(t) := [\phi_t(x), \phi_t(y)].$$

We show that $F(t) = G(t)$.

First,

$$F'(t) = d(\phi_t([x, y])) = \phi_t(d([x, y])).$$

Since d is a derivation,

$$d([x, y]) = [d(x), y] + [x, d(y)].$$

Therefore

$$F'(t) = \phi_t([d(x), y] + [x, d(y)]) = [\phi_t(d(x)), \phi_t(y)] + [\phi_t(x), \phi_t(d(y))].$$

On the other hand,

$$G'(t) = [\phi'_t(x), \phi_t(y)] + [\phi_t(x), \phi'_t(y)] = [\phi_t(d(x)), \phi_t(y)] + [\phi_t(x), \phi_t(d(y))].$$

Hence

$$F'(t) = G'(t).$$

Also,

$$F(0) = [x, y] = G(0).$$

It follows that $F(t) = G(t)$ for all t , and in particular at $t = 1$ we get

$$\exp(d)([x, y]) = [\exp(d)(x), \exp(d)(y)].$$

Thus $\exp(d)$ is a Lie algebra automorphism of L . □

1.2.4 They come from brackets

Definition 1.2.25 (Poisson bracket). *Let A be a commutative associative algebra over a field $\mathbb{k}$. A Poisson bracket on A is a bilinear map*

$$\{\cdot, \cdot\} : A \times A \rightarrow A$$

such that for all $f, g, h \in A$ the following conditions hold:

1. **Skew-symmetry:**

$$\{f, g\} = -\{g, f\}.$$

2. **Jacobi identity:**

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0.$$

3. **Leibniz rule:**

$$\{f, gh\} = \{f, g\}h + g\{f, h\}.$$

In other words, $(A, \{\cdot, \cdot\})$ is a Lie algebra, and for each fixed $f \in A$, the map

$$\text{ad}_f : A \rightarrow A, \quad \text{ad}_f(g) = \{f, g\},$$

is a derivation of the algebra A . An algebra A equipped with such a bracket is called a Poisson algebra.

Remark 1.2.26. *The Leibniz rule in the second variable follows automatically from skew-symmetry:*

$$\{fg, h\} = f\{g, h\} + g\{f, h\}.$$

Thus a Poisson bracket is precisely a Lie bracket compatible with the commutative multiplication on A by the derivation property.

Example 1.2.27 (Canonical Poisson bracket on $\mathbb{R}^{2n}$). Let

$$A = C^\infty(\mathbb{R}^{2n}), \quad (q_1, \dots, q_n, p_1, \dots, p_n)$$

be the standard coordinates. Define

$$\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right).$$

Then $\{\cdot, \cdot\}$ is a Poisson bracket on A . In particular,

$$\{q_i, q_j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}.$$

Definition 1.2.28. Let F be a field of characteristic 0, and let $\mathcal{A}$ be an algebra of differentiable functions in one variable x . For $f, g \in \mathcal{A}$, define

$$[f, g] := fg' - f'g,$$

where $f' = \frac{df}{dx}$ and $g' = \frac{dg}{dx}$.

Definition 1.2.29. To each $f \in \mathcal{A}$, associate the vector field

$$X_f := f(x) \frac{d}{dx}.$$

Proposition 1.2.30. For $f, g \in \mathcal{A}$, one has

$$[X_f, X_g] = X_{[f, g]}.$$

Equivalently,

$$[X_f, X_g] = (fg' - gf') \frac{d}{dx}.$$

Proof. For any test function u , we have

$$X_f(u) = fu', \quad X_g(u) = gu'.$$

Thus

$$X_f(X_g(u)) = f(gu')' = fg'u' + fg u'',$$

and similarly

$$X_g(X_f(u)) = g(fu')' = gf'u' + gf u''.$$

Subtracting, we obtain

$$[X_f, X_g](u) = X_f(X_g(u)) - X_g(X_f(u)) = (fg' - gf')u'.$$

Therefore

$$[X_f, X_g] = (fg' - gf') \frac{d}{dx} = X_{[f,g]}.$$

□

Proposition 1.2.31. *The operation*

$$[f, g] = fg' - f'g$$

is bilinear and skew-symmetric.

Proof. Bilinearity follows immediately from the linearity of differentiation. Also,

$$[g, f] = gf' - g'f = -(fg' - f'g) = -[f, g].$$

Hence the bracket is skew-symmetric.

□

Theorem 1.2.32. *The bracket*

$$[f, g] = fg' - f'g$$

defines a Lie algebra structure on $\mathcal{A}$.

Proof. By the previous proposition, it remains only to verify the Jacobi identity. This follows from the fact that the map

$$f \mapsto X_f = f \frac{d}{dx}$$

identifies this bracket with the usual Lie bracket of vector fields on the line, and the latter always satisfies the Jacobi identity. Hence

$$[f, [g, h]] + [g, [h, f]] + [h, [f, g]] = 0$$

for all $f, g, h \in \mathcal{A}$.

□

1.2.5 They come from Lie groups

Definition 1.2.33. *Let G be a Lie group with identity element e . The Lie algebra of G , denoted by $\text{Lie}(G)$ or $\mathfrak{g}$, is the tangent space*

$$\mathfrak{g} := T_e G$$

equipped with the bracket induced by left-invariant vector fields.

Definition 1.2.34. For each $X \in T_e G$, the corresponding left-invariant vector field $\tilde{X}$ on G is defined by

$$\tilde{X}_g := (dL_g)_e(X), \quad g \in G,$$

where $L_g : G \rightarrow G$ is left translation, $L_g(h) = gh$.

Proposition 1.2.35. The assignment $X \mapsto \tilde{X}$ gives a bijection between $T_e G$ and the space of left-invariant vector fields on G .

Definition 1.2.36. For $X, Y \in \mathfrak{g} = T_e G$, define

$$[X, Y] := [\tilde{X}, \tilde{Y}](e),$$

where on the right-hand side $[\tilde{X}, \tilde{Y}]$ is the usual Lie bracket of vector fields on G .

Proposition 1.2.37. The bracket defined above makes $T_e G$ into a Lie algebra. In particular, it is bilinear, alternating, and satisfies the Jacobi identity.

Example 1.2.38. If $G = GL_n(F)$, where $F = \mathbb{R}$ or $\mathbb{C}$, then

$$\text{Lie}(G) = T_I GL_n(F) \cong M_n(F),$$

and the Lie bracket is

$$[X, Y] = XY - YX.$$

Hence

$$\text{Lie}(GL_n(F)) = \mathfrak{gl}_n(F).$$

Example 1.2.39. If $G = SL_n(F)$, then

$$\text{Lie}(SL_n(F)) = \mathfrak{sl}_n(F) = \{X \in M_n(F) \mid \text{tr}(X) = 0\}.$$

Indeed, differentiating the defining equation $\det(g) = 1$ at the identity gives the trace-zero condition.

Example 1.2.40. If $G = O_n(\mathbb{R})$, then

$$\text{Lie}(O_n(\mathbb{R})) = \mathfrak{o}_n(\mathbb{R}) = \{X \in M_n(\mathbb{R}) \mid X^T = -X\}.$$

This follows by differentiating the relation $g^T g = I$ at the identity.

1.3 Basic concepts of Lie algebra

Let L be a Lie algebra over a field F .

Definition 1.3.1. The derived series of L is defined inductively by

$$L^{(0)} = L, \quad L^{(1)} = [L, L], \quad L^{(i)} = [L^{(i-1)}, L^{(i-1)}] \quad (i \geq 1).$$

The Lie algebra L is called solvable if

$$L^{(n)} = 0$$

for some integer $n \geq 0$.

Definition 1.3.2. The descending central series of L is defined by

$$L^1 = L, \quad L^2 = [L, L], \quad L^i = [L, L^{i-1}] \quad (i \geq 2).$$

The Lie algebra L is called nilpotent if

$$L^n = 0$$

for some integer $n \geq 1$.

Proposition 1.3.3. If L is nilpotent, then L is solvable.

Proof. It is enough to show that

$$L^{(i)} \subseteq L^{2^i} \quad (i \geq 0).$$

We argue by induction on i . The case $i = 0$ is clear. If

$$L^{(i)} \subseteq L^{2^i},$$

then

$$L^{(i+1)} = [L^{(i)}, L^{(i)}] \subseteq [L^{2^i}, L^{2^i}] \subseteq L^{2^{i+1}}.$$

Hence the claim follows for all i . If $L^n = 0$ for some n , then for i sufficiently large one has $2^i \geq n$, so

$$L^{(i)} \subseteq L^{2^i} = 0.$$

Therefore L is solvable. □

Definition 1.3.4. A solvable ideal of L is an ideal which is solvable as a Lie algebra in its own right. Since the sum of two solvable ideals is again solvable, there exists a

unique maximal solvable ideal of L , called the radical of L , denoted by

$$\text{Rad}(L).$$

The Lie algebra L is called semisimple if

$$\text{Rad}(L) = 0.$$

Remark 1.3.5. Thus the three notions fit into the chain

$$\text{nilpotent} \implies \text{solvable},$$

while semisimplicity means that L has no nonzero solvable ideals. In particular, a semisimple Lie algebra cannot contain any nonzero abelian ideal.

Example 1.3.6. Any abelian Lie algebra is nilpotent, hence solvable. On the other hand, the Lie algebra of upper triangular matrices is solvable in general, but need not be nilpotent.

Example 1.3.7. Let

$$L = \left\{ \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \mid a, b \in F \right\} \subseteq \mathfrak{gl}_2(F),$$

Then L is solvable but not nilpotent.

Indeed, if

$$x = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}, \quad y = \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix},$$

then

$$[x, y] = \begin{pmatrix} 0 & ad - bc \\ 0 & 0 \end{pmatrix}.$$

Hence

$$[L, L] = \left\{ \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \mid t \in F \right\}, \quad [[L, L], [L, L]] = 0.$$

Thus the derived series terminates:

$$L^{(1)} = [L, L], \quad L^{(2)} = [L^{(1)}, L^{(1)}] = 0,$$

so L is solvable.

On the other hand, L is not nilpotent, since its descending central series does not reach 0. In fact,

$$L^2 = [L, L] = \left\{ \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \mid t \in F \right\},$$

and for any

$$x = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \in L, \quad y = \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \in L^2,$$

we have

$$[x, y] = \begin{pmatrix} 0 & at \\ 0 & 0 \end{pmatrix} \in L^2.$$

Therefore

$$L^3 = [L, L^2] = L^2,$$

and inductively

$$L^n = L^2 \neq 0 \quad (n \geq 2).$$

So L is not nilpotent.

Chapter 2

Lecture 2 Engel Theorem

2.1 Engel Theorem the first type

Definition 2.1.1. Let L be a Lie algebra over a field F . An enveloping algebra of L is an associative F -algebra A together with a Lie algebra homomorphism

$$\iota : L \rightarrow A^-,$$

where A^- denotes the Lie algebra obtained from A by equipping the underlying vector space of A with the bracket

$$[x, y] = xy - yx,$$

such that the associative algebra A is generated by $\iota(L)$ (or just writing $A = \langle L \rangle_{\text{assoc}}$ or simply $A = \langle L \rangle$).

If, in addition, the pair (A, ι) satisfies the following universal property: for every associative F -algebra B and every Lie algebra homomorphism

$$\varphi : L \rightarrow B^-,$$

there exists a unique associative algebra homomorphism

$$\tilde{\varphi} : A \rightarrow B$$

such that

$$\varphi = \tilde{\varphi} \circ \iota,$$

then A is called the universal enveloping algebra of L .

Theorem 2.1.2. Let A be a finite dimensional enveloping algebra of L , i.e. $L \subseteq A^{(-)}$, $A = \langle L \rangle$. If $\exists n \geq 1$ such that $\forall a \in L, a^n = 0$. Then A is nilpotent.

2.2 Engel Theorem the second type

In fact we can prove a stronger version.

Definition 2.2.1. A subset $S \subseteq L$ is called a Lie subset if

$$[a, b] \in S \quad \text{for all } a, b \in S.$$

Note that a Lie subset need not be a subspace, and hence need not be a Lie subalgebra.

Theorem 2.2.2. Let A be a finite-dimensional associative algebra over F , and let

$$L \subseteq A^{(-)}$$

be a Lie subalgebra such that A is generated by L as an associative algebra. Suppose that

$$L = \text{span}(S)$$

for some Lie subset $S \subseteq L$, and that every element of S is nilpotent in A . Then A is a nilpotent associative algebra.

Remark 2.2.3. In the finite-dimensional situation, the existence of a uniform exponent n is automatic once every element of S is nilpotent in A . Indeed, let

$$m = \dim_F A,$$

and let $a \in A$ be nilpotent. Consider the left multiplication operator

$$L_a : A \rightarrow A, \quad x \mapsto ax.$$

Since a is nilpotent, L_a is a nilpotent linear transformation on the m -dimensional vector space A . Hence

$$L_a^m = 0.$$

But

$$L_a^m = L_{a^m},$$

so $a^m = 0$. Therefore every nilpotent element of A has nilpotency index at most $\dim_F A$. In particular, if every element of S is nilpotent in A , then one may take

$$n = \dim_F A$$

and obtain

$$a^n = 0 \quad \text{for all } a \in S.$$

Thus, in the theorem above, the hypothesis that every element of S is nilpotent in A already implies the existence of a common exponent n .

Theorem 2.2.4 (Zorn's Lemma). *Let $(P, \leq)$ be a partially ordered set. Assume that every chain in P has an upper bound in P . Then P contains a maximal element.*

Proof. We prove the result assuming the Well-Ordering Theorem.

Suppose, for contradiction, that P has no maximal element. Then for every $x \in P$ there exists some $y \in P$ such that

$$x < y.$$

By the Well-Ordering Theorem, we may choose a well-ordering $\preceq$ on P . For each $x \in P$, let $f(x)$ be the $\preceq$ -least element of the set

$$\{y \in P : x < y\}.$$

Then

$$x < f(x) \quad \text{for all } x \in P.$$

We now define, by transfinite recursion, a family (x_α) indexed by ordinals. Suppose that for all $\beta < \alpha$ the elements x_β have been defined and form a chain. By hypothesis, the chain

$$C_\alpha := \{x_\beta : \beta < \alpha\}$$

has an upper bound $u_\alpha \in P$. Define

$$x_\alpha := f(u_\alpha).$$

Since u_α is an upper bound of C_α , for every $\beta < \alpha$ we have

$$x_\beta \leq u_\alpha < x_\alpha.$$

Hence (x_α) is a strictly increasing family:

$$\beta < \alpha \implies x_\beta < x_\alpha.$$

Therefore the map

$$\alpha \longmapsto x_\alpha$$

is injective from the class of all ordinals into the set P . This is impossible, since there is no injection from the class of all ordinals into a set.

This contradiction shows that P must contain a maximal element. $\square$

Lemma 2.2.5. *Let*

$$\mathcal{P} = \{T \subseteq S \mid T \text{ is a Lie subset and } a^n = 0 \text{ for all } a \in T\},$$

partially ordered by inclusion. Then $\mathcal{P}$ has a maximal element.

Proof. We apply Zorn's lemma. Let $\{T_i\}_{i \in I}$ be a chain in $\mathcal{P}$, and set

$$T = \bigcup_{i \in I} T_i.$$

We claim that $T \in \mathcal{P}$.

First, T is a Lie subset. Indeed, if $a, b \in T$, then $a \in T_i$ and $b \in T_j$ for some $i, j \in I$. Since $\{T_i\}_{i \in I}$ is a chain, one of T_i, T_j is contained in the other; say $T_i \subseteq T_j$. Then $a, b \in T_j$, and since T_j is a Lie subset,

$$[a, b] \in T_j \subseteq T.$$

Second, if $a \in T$, then $a \in T_i$ for some $i \in I$, so $a^n = 0$ because $T_i \in \mathcal{P}$.

Thus $T \in \mathcal{P}$, and clearly T is an upper bound of the chain. Therefore every chain in $\mathcal{P}$ has an upper bound. By Zorn's lemma, $\mathcal{P}$ has a maximal element. $\square$

Proof of the theorem. Consider the collection

$$\mathcal{P} = \{T \subseteq S \mid T \text{ is a Lie subset of } L \text{ and } \langle T \rangle \text{ is a nilpotent associative subalgebra of } A\},$$

partially ordered by inclusion. Here $\langle T \rangle$ denotes the associative subalgebra of A generated by T .

We first show that every chain in $\mathcal{P}$ has an upper bound. Let $\{T_i\}_{i \in I}$ be a chain in $\mathcal{P}$, and set

$$T = \bigcup_{i \in I} T_i.$$

Since the T_i form a chain, T is again a Lie subset of L . Let $N = \dim_F A$. Since each $\langle T_i \rangle$ is a nilpotent subalgebra of the finite-dimensional algebra A , we have

$$\langle T_i \rangle^N = 0 \quad \text{for all } i \in I.$$

Now take any $x_1, \dots, x_N \in \langle T \rangle$. Each x_j belongs to $\langle T_{i_j} \rangle$ for some $i_j \in I$. Since the family $\{T_i\}$ is totally ordered, there exists $k \in I$ such that

$$T_{i_1} \cup \dots \cup T_{i_N} \subseteq T_k.$$

Hence

$$x_1, \dots, x_N \in \langle T_k \rangle,$$

and therefore

$$x_1 \cdots x_N = 0.$$

Thus $\langle T \rangle^N = 0$, so $T \in \mathcal{P}$. By Zorn's lemma, $\mathcal{P}$ has a maximal element, say S_1 .

Set

$$B = \langle S_1 \rangle.$$

Then B is nilpotent, so there exists $t \geq 1$ such that

$$B^t = 0.$$

We claim that if $S_1 \neq S$, then there exists $s \in S \setminus S_1$ such that

$$[S_1, s] \subseteq S_1.$$

Assume the contrary. Choose $s_1 \in S \setminus S_1$. By assumption, there exists $x_1 \in S_1$ such that

$$s_2 = [x_1, s_1] \notin S_1.$$

Since S is a Lie subset, $s_2 \in S$. Repeating this argument, we construct inductively elements

$$x_1, x_2, \dots \in S_1, \quad s_1, s_2, \dots \in S \setminus S_1$$

such that

$$s_{r+1} = [x_r, s_r] \quad (r \geq 1).$$

Thus

$$s_{r+1} = \text{ad}(x_r) \cdots \text{ad}(x_1)(s_1).$$

We now show that this process cannot continue for $2t$ steps. Indeed, for any $y_1, \dots, y_r \in B$, the iterated commutator

$$\text{ad}(y_r) \cdots \text{ad}(y_1)(s_1)$$

is a sum of terms of the form

$$\pm u s_1 v,$$

where $u, v \in B$ and the total number of factors from B occurring in u and v is r . Therefore, if $r \geq 2t$, then in each such term either $u \in B^t$ or $v \in B^t$, so the term is 0. Hence

$$\text{ad}(y_r) \cdots \text{ad}(y_1)(s_1) = 0 \quad \text{for all } r \geq 2t.$$

Applying this with $y_i = x_i \in S_1 \subseteq B$, we obtain

$$s_{2t+1} = 0,$$

which is impossible since $0 \in S_1$. This contradiction proves the claim.

Choose $s \in S \setminus S_1$ such that

$$[S_1, s] \subseteq S_1.$$

Let

$$S_2 = S_1 \cup \{s\}.$$

Then S_2 is a Lie subset of L : indeed, brackets of elements of S_1 remain in S_1 , $[S_1, s] \subseteq S_1$, $[s, S_1] \subseteq S_1$ by antisymmetry, and $[s, s] = 0 \in S_1$.

We next show that $\langle S_2 \rangle$ is nilpotent. Since $[S_1, s] \subseteq S_1$, we have

$$[B, s] \subseteq B.$$

Hence for every $b \in B$,

$$bs = sb + [b, s], \quad [b, s] \in B.$$

By repeated use of this identity, every product in the algebra generated by B and s can be written as a linear combination of terms of the form

$$s^r b, \quad b \in B,$$

with $r \geq 0$. Since $s^m = 0$, we may assume $0 \leq r \leq m - 1$.

Now consider a product of length mt in $\langle S_2 \rangle$. After rewriting it as above, each term has the form $s^r b$ with $0 \leq r \leq m - 1$. Since the original length is mt , such a term must contain at least

$$mt - r \geq mt - (m - 1) \geq t$$

factors coming from B . Therefore $b \in B^t = 0$, and every such term is zero. It follows that

$$\langle S_2 \rangle^{mt} = 0.$$

Thus $\langle S_2 \rangle$ is nilpotent.

But $S_1 \subsetneq S_2$ and $S_2 \in \mathcal{P}$, contradicting the maximality of S_1 . Hence $S_1 = S$. Therefore

$$A = \langle L \rangle = \langle S \rangle = \langle S_1 \rangle$$

is nilpotent. □

2.3 Engel Theorem the third type

Definition 2.3.1. Let L be a Lie algebra over a field F . An L -module is a vector space V over F together with a bilinear map

$$L \times V \rightarrow V, \quad (x, v) \mapsto x \cdot v,$$

such that

$$[x, y] \cdot v = x \cdot (y \cdot v) - y \cdot (x \cdot v)$$

for all $x, y \in L$ and $v \in V$. In other words, V is an L -module if and only if there exists a Lie algebra homomorphism

$$\phi : L \rightarrow \mathfrak{gl}(V).$$

such that

$$\phi(x)(v) = x \cdot v$$

for all $x \in L$ and $v \in V$.

Proposition 2.3.2. *Let L be a Lie algebra over a field F , and let M be a vector space over F . Suppose a bilinear operation*

$$L \times M \rightarrow M, \quad (a, m) \mapsto a \cdot m$$

is given. Then the following conditions are equivalent:

1. *The vector space $L \oplus M$ becomes a Lie algebra under the bracket defined by*

$$[a + m, b + n] = [a, b] + a \cdot n - b \cdot m \quad (a, b \in L, m, n \in M),$$

equivalently,

$$[a, m] = a \cdot m, \quad [m, a] = -a \cdot m$$

for all $a \in L, m \in M$, and

$$[m, n] = 0$$

for all $m, n \in M$.

2. *For all $a, b \in L$ and $m \in M$,*

$$[a, b] \cdot m = a \cdot (b \cdot m) - b \cdot (a \cdot m).$$

3. *The map*

$$\rho : L \rightarrow \mathfrak{gl}_F(M), \quad \rho(a)(m) = a \cdot m,$$

is a Lie algebra homomorphism.

When these equivalent conditions hold, we say that M is an L -module.

Proof. We prove $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$.

$(1) \Rightarrow (2)$. Assume that $L \oplus M$ is a Lie algebra with the stated bracket. Since every Lie algebra satisfies the Jacobi identity, for any $a, b \in L$ and $m \in M$ we have

$$[a, [b, m]] + [b, [m, a]] + [m, [a, b]] = 0.$$

Now

$$[b, m] = b \cdot m, \quad [m, a] = -a \cdot m, \quad [a, b] \in L.$$

Because M is abelian, this gives

$$[a, b \cdot m] + [b, -a \cdot m] + [m, [a, b]] = 0.$$

Using again the definition of the bracket,

$$a \cdot (b \cdot m) - b \cdot (a \cdot m) - [a, b] \cdot m = 0.$$

Hence

$$[a, b] \cdot m = a \cdot (b \cdot m) - b \cdot (a \cdot m).$$

(2) $\Rightarrow$ (3). Define

$$\rho : L \rightarrow \mathfrak{gl}_F(M), \quad \rho(a)(m) = a \cdot m.$$

Since the action is bilinear, ρ is linear. For any $a, b \in L$ and $m \in M$, condition (2) gives

$$\rho([a, b])(m) = [a, b] \cdot m = a \cdot (b \cdot m) - b \cdot (a \cdot m) = \rho(a)\rho(b)(m) - \rho(b)\rho(a)(m).$$

Therefore

$$\rho([a, b]) = [\rho(a), \rho(b)],$$

so ρ is a Lie algebra homomorphism.

(3) $\Rightarrow$ (1). Assume that $\rho : L \rightarrow \mathfrak{gl}_F(M)$ is a Lie algebra homomorphism, and define a bracket on $L \oplus M$ by

$$[a + m, b + n] = [a, b] + \rho(a)(n) - \rho(b)(m).$$

This bracket is bilinear and skew-symmetric. It remains to verify the Jacobi identity.

Take $a, b, c \in L$ and $m, n, p \in M$. Write

$$x = a + m, \quad y = b + n, \quad z = c + p.$$

A direct calculation shows that the L -component of

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]]$$

is

$$[a, [b, c]] + [b, [c, a]] + [c, [a, b]],$$

which is zero because L is a Lie algebra.

For the M -component, expanding the definition of the bracket yields

$$\begin{aligned} & \rho(a)\rho(b)(p) - \rho(a)\rho(c)(n) - \rho([b, c])(m) + \rho(b)\rho(c)(m) - \rho(b)\rho(a)(p) - \rho([c, a])(n) \\ & + \rho(c)\rho(a)(n) - \rho(c)\rho(b)(m) - \rho([a, b])(p). \end{aligned}$$

Grouping the terms by m, n, p , we obtain

$$\begin{aligned} & (\rho(b)\rho(c) - \rho(c)\rho(b) - \rho([b, c]))(m) \\ & + (\rho(c)\rho(a) - \rho(a)\rho(c) - \rho([c, a]))(n) \\ & + (\rho(a)\rho(b) - \rho(b)\rho(a) - \rho([a, b]))(p). \end{aligned}$$

Since ρ is a Lie algebra homomorphism, each bracketed expression is zero. Hence the M -component also vanishes, so the Jacobi identity holds.

Therefore $L \oplus M$ is a Lie algebra with the stated bracket.

This proves that the three conditions are equivalent. $\square$

Theorem 2.3.3. *Let L be a finite dimensional Lie algebra over a field F . If $\forall a \in L$, $\text{ad}(a)$ is nilpotent, then L is nilpotent.*

Proof. Note that since L is finite-dimensional, say $\dim L = n < \infty$, and for every $a \in L$ we have

$$(\text{ad } a)^n = 0.$$

Hence we may apply the previous theorem: if for all $a \in L$,

$$(\text{ad } a)^n = 0,$$

then

$$(\text{ad } L)^n = 0,$$

and therefore

$$L^{n+1} = 0.$$

So L is nilpotent. $\square$

Remark 2.3.4. *If L is infinite-dimensional, one should distinguish two cases:*

$$\begin{cases} \text{ad } a \text{ nilpotent with a uniform bound } n \implies L \text{ nilpotent,} \\ \text{no uniform bound on the nilpotency indices} \not\Rightarrow L \text{ nilpotent.} \end{cases}$$

Remark 2.3.5. *There are three related versions, with*

$$\text{type (2)} \implies \text{type (1)} \implies \text{type (3)}.$$

Type (2): Let $S \subseteq L$ be a Lie subset. And S generates L as a Lie algebra. Assume there exists n such that for every $a \in S$

$$a^n = 0.$$

Then $A = \langle L \rangle$ is a nilpotent algebra. (When A is a finite-dimensional associative algebra).

In particular, if we choose S to be L , then we get type (1).

Type (1): Let A be a finite dimensional enveloping algebra of L , i.e. $L \subseteq A^{(-)}$, $A = \langle L \rangle$. If $\exists n \geq 1$ such that $\forall a \in L$, $a^n = 0$. Then A is nilpotent.

And consider A as $\text{ad}(L)$, we get type (3).

Type (3): Let L be a finite dimensional Lie algebra over a field F . If $\forall a \in L$, $\text{ad}(a)$ is nilpotent, then L is nilpotent.

We also have the following truth:

Corollary 2.3.6. *Let M be an L -module with $\dim M < \infty$, that is, let*

$$\varphi : L \rightarrow \mathfrak{gl}_F(M)$$

be a representation. Assume that for every $a \in L$, the operator $\varphi(a)$ is nilpotent. Equivalently, for every $a \in L$ there exists n such that

$$\underbrace{a(a(\cdots(aM)\cdots))}_{n \text{ times}} = 0.$$

Then there exists N such that

$$\underbrace{L \cdots L}_N M = 0.$$

In other words, if every $\varphi(a)$ is nilpotent, then the Lie algebra $\varphi(L)$ is nilpotent. Hence there exists a basis of M in which every element of $\varphi(L)$ is represented by a strictly upper triangular matrix:

$$\varphi(L) \subseteq \left\{ \begin{pmatrix} 0 & * & \cdots & * \\ 0 & 0 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix} \right\}.$$

Remark 2.3.7. *This corollary is the module-theoretic form of Engel's theorem. Indeed, the representation*

$$\varphi : L \rightarrow \mathfrak{gl}_F(M)$$

has image $\varphi(L)$, which is a Lie subalgebra of $\mathfrak{gl}_F(M)$. By assumption, every element of $\varphi(L)$ is a nilpotent endomorphism of the finite-dimensional vector space M . Hence, by Engel's theorem, the Lie algebra $\varphi(L)$ is nilpotent.

The conclusion

$$\underbrace{L \cdots L}_N M = 0$$

means that there exists a uniform integer N such that for all $a_1, \dots, a_N \in L$ and all $m \in M$,

$$a_1(a_2(\cdots(a_N m)\cdots)) = 0.$$

Thus the action is nilpotent not only for each fixed $a \in L$, but uniformly for all sufficiently long iterated actions.

A further consequence of Engel's theorem is that there exists a basis of M with respect to which every operator $\varphi(a)$ is represented by a strictly upper triangular matrix. Equivalently, M admits a chain of subspaces

$$0 = M_0 \subset M_1 \subset \cdots \subset M_r = M, \quad \dim M_i = i,$$

such that

$$L \cdot M_i \subseteq M_{i-1} \quad (1 \leq i \leq r).$$

Explanation: By Engel's theorem, $\varphi(L)$ is nilpotent. Hence

$$\underbrace{L \cdots L}_N M = 0$$

Take the minimal N such that $L^N M = 0$ and $L^{N-1} M \neq 0$. Then we have a non-zero vector v_1 in $L^{N-1} M$ which is killed by $\varphi(L)$. Denote $M_1 = \text{span}(v_1)$. Then $L \cdot M_1 \subseteq M_0$. Now consider M/M_1 . By Engel's theorem and by induction hypothesis, $\varphi(L)$ is nilpotent on M/M_1 . Hence there exists a basis of M/M_1 with respect to which every element of $\varphi(L)$ is represented by a strictly upper triangular matrix. Equivalently, M/M_1 admits a chain of subspaces

$$0 = M_1/M_1 \subset M_2/M_1 \subset \cdots \subset M_r/M_1 = M/M_1, \quad \dim M_i/M_1 = i,$$

such that

$$L \cdot (M_i/M_1) \subseteq M_{i-1}/M_1 \quad (1 \leq i \leq r).$$

Now we can take a basis of M_1 and M/M_1 to get a basis of M with respect to which every element of $\varphi(L)$ is represented by a strictly upper triangular matrix.

In particular, this corollary reduces to the usual form of Engel's theorem when one takes $M = L$ and $\varphi = \text{ad}$.

Remark 2.3.8. According to this corollary and type (2), we can rewrite it as:

Let M be an L -module with $\dim M < \infty$, that is, let

$$\varphi : L \rightarrow \mathfrak{gl}_F(M)$$

be a representation. Assume that for every $a \in S$, where S is a Lie subset of L and $\langle S \rangle = L$, the operator $\varphi(a)$ is nilpotent. Equivalently, for every $a \in S$ there exists n such that

$$\underbrace{a(a \cdots (a M) \cdots)}_{n \text{ times}} = 0.$$

Then there exists N such that

$$\underbrace{L \cdots L}_N M = 0.$$

For this result, we do not need L or the enveloping algebra $A = \langle L \rangle$ to be finite-dimensional now, we only need such n is a universal bound.

Chapter 3

Lecture 3: Lie's theorem and Jordan-Chevalley decomposition

3.1 Lie's theorem

Lemma 3.1.1. *Let F be a field with $\text{char } F = 0$ or $\text{char } F > n$. Let $a, b \in M_n(F)$ satisfy*

$$[a, [a, b]] = 0.$$

Then $[a, b]$ is nilpotent. In fact,

$$[a, b]^n = 0.$$

Proof. Set

$$\delta := \text{ad}(a), \quad \delta(x) = [a, x].$$

Since δ is an inner derivation of the associative algebra $M_n(F)$, we have

$$\delta(xy) = \delta(x)y + x\delta(y) \quad \text{for all } x, y \in M_n(F).$$

Our assumption is precisely

$$\delta^2(b) = 0.$$

Now let us prove the following two facts:

Claim 1. For every $m \geq 1$ and every $r > m$,

$$\text{End } \delta^r(b^m) = 0.$$

Indeed, each application of δ to the product b^m differentiates one of the m factors. Since $\delta^2(b) = 0$, no factor can be differentiated twice. Hence after more than m applications, every term vanishes.

Claim 2. For every $m \geq 1$,

$$\delta^m(b^m) = m! \delta(b)^m = m! [a, b]^m.$$

We prove this by induction on m . The case $m = 1$ is clear. Assume the formula holds for $m - 1$. Using the derivation rule,

$$\delta(b^m) = \delta(b)b^{m-1} + b\delta(b^{m-1}).$$

Applying δ^{m-1} and using Leibniz repeatedly, together with $\delta^2(b) = 0$, one finds that the only surviving terms are those in which each of the m copies of b is hit exactly once. There are exactly $m!$ such terms, and each contributes

$$\delta(b)^m.$$

Hence

$$\delta^m(b^m) = m! \delta(b)^m.$$

Now apply Cayley–Hamilton to b :

$$b^n + \beta_{n-1}b^{n-1} + \cdots + \beta_1b + \beta_0I = 0.$$

Apply δ^n to both sides. By Claim 1, for every $i < n$ we have

$$\delta^n(b^i) = 0,$$

and also $\delta^n(I) = 0$. Therefore

$$\delta^n(b^n) = 0.$$

By Claim 2 with $m = n$, this becomes

$$n! [a, b]^n = 0.$$

Since $\text{char } F = 0$ or $\text{char } F > n$, we have $n! \neq 0$ in F . Hence

$$[a, b]^n = 0.$$

Therefore $[a, b]$ is nilpotent. □

Theorem 3.1.2 (Lie’s theorem). *Assume that $\text{char } F = 0$ or $\text{char } F > \dim_F M$. F is algebraically closed. Let L be a solvable Lie algebra and let M be an L -module with*

$$\dim_F M = n < \infty.$$

Then there exists a basis of M such that, relative to this basis, the image of L (from here when talk about L acting on its module M , we simply use L instead of $\varphi(L)$)

consists of upper triangular matrices; that is,

$$L \subseteq \left\{ \begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & * \end{pmatrix} \right\}.$$

Proof. We argue first that M has a common eigenvector for the action of L .

Assume first that M is irreducible as an L -module. Since we are mainly concerned with the image of

$$L \rightarrow \mathfrak{gl}_F(M),$$

we may replace L by its image and assume that $L \subseteq \mathfrak{gl}_F(M)$.

Because L is solvable, its derived series

$$L \supseteq L^{[1]} \supseteq L^{[2]} \supseteq \cdots$$

terminates at 0. Hence there exists an integer $s \geq 0$ such that

$$L^{[s]} \neq 0, \quad L^{[s+1]} = [L^{[s]}, L^{[s]}] = 0.$$

If $s = 0$, then L is abelian, and since F is algebraically closed, there exists an eigenvector $v \in M$ for some $x \in L$. As L is abelian, every element of L preserves the eigenspace of x , and by irreducibility this gives a common eigenvector for all $x \in L$.

Now assume $s \geq 1$. Choose $a \in L^{[s]}$ and $b \in L$. Then

$$[a, [a, b]] \in [L^{[s]}, L^{[s]}] = L^{[s+1]} = 0,$$

so

$$(\text{ad } a)^2(b) = 0.$$

Consider the set

$$S = \{[a, b] \mid a \in L^{[s]}, b \in L\}.$$

This is a Lie subset of L , and every element of S is nilpotent in $\mathfrak{gl}_F(M)$. By Engel's theorem,

$$\text{span}_F(S) = [L^{[s]}, L]$$

is a nilpotent Lie algebra of linear transformations on M .

Now let I be any ideal of L . Then IM is an L -submodule of M , because for $x \in L$, $y \in I$, and $m \in M$,

$$x \cdot (y \cdot m) = y \cdot (x \cdot m) + [x, y] \cdot m \in IM,$$

since $[x, y] \in I$. As M is irreducible, it follows that

$$IM = 0 \quad \text{or} \quad IM = M.$$

Applying this to the ideal $I = L^{[s]}$, we cannot have $L^{[s]}M = M$, because $L^{[s]}$ acts nilpotently on M . Hence

$$L^{[s]}M = 0,$$

that is, $L^{[s]}$ acts trivially on M .

Thus every element of $L^{[s]}$ commutes with the action of L on M . In particular, for any $c \in L^{[s-1]}$ and $d \in L^{[s-1]}$,

$$[c, d] \in L^{[s]}$$

acts as 0 on M . Therefore $L^{[s]} = 0$ in the image of L in $\mathfrak{gl}_F(M)$, contrary to the choice of s . This contradiction shows that $s = 0$, so L is abelian (on M), and hence M contains a common eigenvector for all elements of L .

Now let $0 \neq v_1 \in M$ be such a common eigenvector. Then Fv_1 is a one-dimensional L -submodule of M . Consider the quotient M/Fv_1 , which is again a finite-dimensional L -module. By induction on $\dim_F M$, there exists a basis of M/Fv_1 with respect to which the induced action of L is upper triangular. Lifting this basis to M and adjoining v_1 , we obtain a basis of M relative to which every element of L is represented by an upper triangular matrix.

Therefore there exists a basis of M such that the image of L consists of upper triangular matrices. $\square$

Proposition 3.1.3. *Let L be a solvable Lie algebra over F with $\text{char } F = 0$ or $\text{char } F > \dim_F M$. F is algebraically closed. Then the followings are equivalent:*

1. *Any irreducible finite dimensional L -module is 1-dimensional.*
2. *If M is a finite dimensional L -module, then there exist a basis of M such that the image of L consists of upper triangular matrices.*
3. *If M is any finite dimensional L -module. Then $[L, L]$ acts nilpotently on M .*

Proof. We prove

$$(2) \Rightarrow (1), \quad (1) \Rightarrow (2), \quad (2) \Rightarrow (3), \quad (3) \Rightarrow (1).$$

$(2) \Rightarrow (1)$. Let M be an irreducible finite-dimensional L -module. By (2), there exists a basis of M relative to which every operator in the image of L is upper triangular. Then the subspace spanned by the first basis vector is stable under L , hence is a nonzero submodule of M . Since M is irreducible, this submodule must be all of M . Therefore $\dim_F M = 1$.

$(1) \Rightarrow (2)$. We argue by induction on $\dim_F M$.

If $\dim_F M = 1$, there is nothing to prove. Assume $\dim_F M > 1$, and suppose the statement is known for all modules of smaller dimension. Since M is finite-dimensional, it contains an irreducible submodule N . By (1), we have $\dim_F N = 1$.

Choose $0 \neq v_1 \in N$, so that $N = Fv_1$. Then N is an L -submodule, and hence M/N is

again a finite-dimensional L -module. By the induction hypothesis, there exists a basis

$$\overline{v_2}, \dots, \overline{v_n}$$

of M/N such that the induced action of L on M/N is given by upper triangular matrices. Choose representatives $v_2, \dots, v_n \in M$ of these cosets. Then

$$v_1, v_2, \dots, v_n$$

is a basis of M .

Since Fv_1 is L -stable, every element of L sends v_1 into Fv_1 . Since the induced action on M/Fv_1 is upper triangular with respect to

$$\overline{v_2}, \dots, \overline{v_n},$$

it follows that the action of L on M is upper triangular with respect to

$$v_1, v_2, \dots, v_n.$$

Thus (2) holds.

(2) $\Rightarrow$ (3). Let M be a finite-dimensional L -module, and choose a basis of M such that every element of the image of L is represented by an upper triangular matrix. Then for any $x, y \in L$,

$$\rho([x, y]) = [\rho(x), \rho(y)].$$

Since the commutator of two upper triangular matrices has zero diagonal entries, $\rho([x, y])$ is strictly upper triangular. Hence every element of $\rho([L, L])$ is strictly upper triangular, and therefore nilpotent. Thus $[L, L]$ acts nilpotently on M .

(3) $\Rightarrow$ (1). Let M be an irreducible finite-dimensional L -module. By (3), every element of $[L, L]$ acts nilpotently on M . Hence the Lie algebra $\rho([L, L]) \subseteq \mathfrak{gl}(M)$ consists entirely of nilpotent endomorphisms. By Engel's theorem, there exists a nonzero vector $v \in M$ such that

$$\rho([L, L])v = 0.$$

Thus $[L, L]$ annihilates v .

Let $N = U(L)v$ be the submodule generated by v . Since $[L, L]v = 0$, the action of L on N factors through the abelian Lie algebra $L/[L, L]$. Therefore N is a finite-dimensional module for the abelian Lie algebra $L/[L, L]$.

Because F is algebraically closed, every irreducible finite-dimensional module over an abelian Lie algebra is 1-dimensional. Hence N contains a 1-dimensional submodule. Since M is irreducible and $0 \neq N \subseteq M$, we have $N = M$. Therefore M itself contains a 1-dimensional submodule, and irreducibility forces $\dim_F M = 1$.

This proves (1).

Therefore

$$(1) \Longleftrightarrow (2) \Longleftrightarrow (3).$$

□

Remark 3.1.4. *The assumption that F is algebraically closed is used in the last step of the proof of (3) $\Rightarrow$ (1), namely in the fact that every irreducible finite-dimensional module over the abelian Lie algebra $L/[L, L]$ is 1-dimensional. Equivalently, one needs the existence of eigenvalues and eigenvectors for commuting linear operators.*

Example 3.1.5. *Assume that $\text{char } F = 2$. Then $\mathfrak{gl}(2, F)$ is solvable. Indeed, every element of $\mathfrak{gl}(2, F)$ has the form*

$$\begin{pmatrix} \alpha & \gamma \\ \beta & \delta \end{pmatrix}, \quad \alpha, \beta, \gamma, \delta \in F.$$

A direct computation shows that the commutator of two such matrices is always of the form

$$\left[\begin{pmatrix} \alpha & \gamma \\ \beta & \delta \end{pmatrix}, \begin{pmatrix} \alpha' & \gamma' \\ \beta' & \delta' \end{pmatrix} \right] = \begin{pmatrix} d & x \\ y & d \end{pmatrix},$$

for some $d, x, y \in F$. Hence

$$\mathfrak{gl}(2, F)^{(1)} = [\mathfrak{gl}(2, F), \mathfrak{gl}(2, F)] \subseteq \left\{ \begin{pmatrix} d & x \\ y & d \end{pmatrix} : d, x, y \in F \right\}.$$

Now if

$$A = \begin{pmatrix} \alpha & \gamma \\ \beta & \alpha \end{pmatrix} \quad (\text{since } \text{char } F = 2),$$

and similarly

$$B = \begin{pmatrix} \alpha' & \gamma' \\ \beta' & \alpha' \end{pmatrix},$$

then their commutator is of the form

$$[A, B] = \begin{pmatrix} d & 0 \\ 0 & d \end{pmatrix}$$

for some $d \in F$. Therefore

$$\mathfrak{gl}(2, F)^{(2)} \subseteq \left\{ \begin{pmatrix} d & 0 \\ 0 & d \end{pmatrix} : d \in F \right\},$$

and so

$$\mathfrak{gl}(2, F)^{(3)} = 0.$$

Thus $\mathfrak{gl}(2, F)$ is solvable.

3.2 Generalized eigenvectors and generalized eigenspaces

Definition 3.2.1. Let V be a finite-dimensional vector space over a field F where F is algebraically closed, and let

$$T \in \text{End}_F(V).$$

Suppose that the characteristic polynomial of T splits as

$$f(x) = (x - \lambda_1)^{a_1} \cdots (x - \lambda_k)^{a_k},$$

where $\lambda_1, \dots, \lambda_k$ are distinct. For each i , define

$$V_{\lambda_i} := \ker(T - \lambda_i I)^{a_i}.$$

This is called the generalized eigenspace of T corresponding to λ_i .

Theorem 3.2.2. With the notation above,

$$V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_k}.$$

Proof. For each i , let

$$P_i(x) = \frac{f(x)}{(x - \lambda_i)^{a_i}} = \prod_{j \neq i} (x - \lambda_j)^{a_j}.$$

Since the polynomials $P_1(x), \dots, P_k(x)$ have greatest common divisor 1, there exist polynomials

$$Q_1(x), \dots, Q_k(x) \in F[x]$$

such that

$$Q_1(x)P_1(x) + \cdots + Q_k(x)P_k(x) = 1.$$

Replacing x by T , we obtain

$$Q_1(T)P_1(T) + \cdots + Q_k(T)P_k(T) = I_V.$$

Hence every $v \in V$ can be written as

$$v = \sum_{i=1}^k Q_i(T)P_i(T)v.$$

Now

$$(T - \lambda_i I)^{a_i} Q_i(T)P_i(T) = Q_i(T)f(T) = 0,$$

because $f(T) = 0$ by the Cayley–Hamilton theorem. Therefore

$$Q_i(T)P_i(T)v \in \ker(T - \lambda_i I)^{a_i} = V_{\lambda_i},$$

and so

$$V = V_{\lambda_1} + \cdots + V_{\lambda_k}.$$

To show that the sum is direct, let

$$v_1 + \cdots + v_k = 0, \quad v_i \in V_{\lambda_i}.$$

Fix i . Since $P_i(\lambda_i) \neq 0$, the polynomial $P_i(x)$ is relatively prime to $(x - \lambda_i)^{a_i}$. Hence there exist polynomials $A_i(x), B_i(x) \in F[x]$ such that

$$A_i(x)P_i(x) + B_i(x)(x - \lambda_i)^{a_i} = 1.$$

Replacing x by T and applying to $v_i \in V_{\lambda_i}$, we get

$$A_i(T)P_i(T)v_i = v_i,$$

because $(T - \lambda_i I)^{a_i}v_i = 0$. Thus $P_i(T)$ is invertible on V_{λ_i} .

On the other hand, if $j \neq i$, then

$$P_i(T)v_j = 0,$$

because $P_i(x)$ contains the factor $(x - \lambda_j)^{a_j}$ and

$$(T - \lambda_j I)^{a_j}v_j = 0.$$

Applying $P_i(T)$ to the relation

$$v_1 + \cdots + v_k = 0,$$

we obtain

$$P_i(T)v_i = 0.$$

Since $P_i(T)$ is invertible on V_{λ_i} , it follows that $v_i = 0$. As i was arbitrary, all v_i are zero. Therefore the sum is direct. $\square$

Chapter 4

Homework

4.1 Homework week 1

Exercise 4.1.1. *Prove the simplicity of Lie algebras $\mathfrak{sl}_n$, $\mathfrak{o}_n$, $\mathfrak{sp}_{2n}$ and $W(n)$.*

I can only prove when base field is of characteristic 0 and $\neq 2$. (Or $\text{Char } F \nmid n$, which is essentially the same. If $\text{Char } F \mid n$ or $\text{Char } F = 2$, then either I will be contained in the center of L , or the alternative becomes symmetry, things could be more complicated).

The simplicity of $\mathfrak{sl}_n$

Proof. Let $I \neq 0$ be an ideal of $\mathfrak{sl}_n(F)$. We prove that $I = \mathfrak{sl}_n(F)$.

Write

$$\mathfrak{sl}_n(F) = \mathfrak{h} \oplus \bigoplus_{i \neq j} F e_{ij},$$

where $\mathfrak{h}$ is the subalgebra of diagonal trace-zero matrices. Choose $0 \neq x \in I$ and write

$$x = y + \sum_{t=1}^s c_t e_{i_t j_t}, \quad y \in \mathfrak{h},$$

where the pairs (i_t, j_t) are distinct and $c_t \neq 0$ for all t .

We first show that I contains some off-diagonal matrix unit e_{kl} .

If $s = 0$, then $x = y \in \mathfrak{h}$ is non-zero. Writing

$$y = \text{diag}(d_1, \dots, d_n),$$

we can choose $k \neq l$ such that $d_k \neq d_l$. Then

$$[y, e_{kl}] = (d_k - d_l)e_{kl} \in I,$$

hence $e_{kl} \in I$.

Assume now that $s \geq 1$. For

$$h = \text{diag}(\lambda_1, \dots, \lambda_n) \in \mathfrak{h},$$

define

$$\alpha_{ij}(h) = \lambda_i - \lambda_j.$$

Since $\text{char } F = 0$, the field F is infinite, so we may choose $h \in \mathfrak{h}$ such that

$$\mu_t := \alpha_{i_t j_t}(h) \neq 0 \quad \text{and} \quad \mu_1, \dots, \mu_s \text{ are pairwise distinct.}$$

Because $\mathfrak{h}$ is abelian, for every $r \geq 1$,

$$(\text{ad } h)^r(x) = \sum_{t=1}^s c_t \mu_t^r e_{i_t j_t}.$$

Put

$$z_r := (\text{ad } h)^r(x) \in I, \quad r = 1, \dots, s.$$

Then

$$\begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_s \end{pmatrix} = \begin{pmatrix} \mu_1 & \mu_2 & \cdots & \mu_s \\ \mu_1^2 & \mu_2^2 & \cdots & \mu_s^2 \\ \vdots & \vdots & \ddots & \vdots \\ \mu_1^s & \mu_2^s & \cdots & \mu_s^s \end{pmatrix} \begin{pmatrix} c_1 e_{i_1 j_1} \\ c_2 e_{i_2 j_2} \\ \vdots \\ c_s e_{i_s j_s} \end{pmatrix}.$$

The coefficient matrix is invertible: it is the product of $\text{diag}(\mu_1, \dots, \mu_s)$ and a Vandermonde matrix in $\mu_1, \dots, \mu_s$. Therefore each $c_t e_{i_t j_t}$ is a linear combination of $z_1, \dots, z_s$, so $e_{i_t j_t} \in I$ for every t . In particular, I contains some e_{kl} with $k \neq l$.

Fix such an element $e_{kl} \in I$. Since I is an ideal,

$$h_{kl} := [e_{kl}, e_{lk}] = e_{kk} - e_{ll} \in I.$$

For every $j \neq k$,

$$[h_{kl}, e_{kj}] = \begin{cases} 2e_{kl}, & j = l, \\ e_{kj}, & j \neq l, \end{cases}$$

so $e_{kj} \in I$ for all $j \neq k$. Similarly, for every $i \neq k$,

$$[e_{ik}, h_{kl}] = \begin{cases} 2e_{lk}, & i = l, \\ e_{ik}, & i \neq l, \end{cases}$$

hence $e_{ik} \in I$ for all $i \neq k$.

Now let $i \neq j$. If $i = k$ or $j = k$, then we already know $e_{ij} \in I$. If $i, j \neq k$, then

$$e_{ij} = [e_{ik}, e_{kj}] \in I.$$

Thus every off-diagonal matrix unit e_{ij} belongs to I .

Finally, for $i \neq j$,

$$[e_{ij}, e_{ji}] = e_{ii} - e_{jj} \in I.$$

These elements span $\mathfrak{h}$, hence $\mathfrak{h} \subseteq I$. Therefore

$$\mathfrak{sl}_n(F) = \mathfrak{h} \oplus \bigoplus_{i \neq j} Fe_{ij} \subseteq I,$$

so $I = \mathfrak{sl}_n(F)$. Hence $\mathfrak{sl}_n(F)$ is simple. $\square$

The simplicity of $\mathfrak{o}_n$ 9‘=’

Proof. Let $\mathfrak{g} = \mathfrak{o}_n(F)$. For $i \neq j$, set

$$X_{ij} = E_{ij} - E_{ji},$$

so $X_{ji} = -X_{ij}$ and $\{X_{ij} \mid 1 \leq i < j \leq n\}$ is a basis of $\mathfrak{g}$. The bracket is

$$[X_{ij}, X_{kl}] = \delta_{jk}X_{il} - \delta_{ik}X_{jl} - \delta_{jl}X_{ik} + \delta_{il}X_{jk}.$$

If $n = 1$ or $n = 2$, then $\mathfrak{g}$ is abelian, hence not simple. For $n = 4$, let

$$I_+ = \text{span}\{X_{12} + X_{34}, X_{13} - X_{24}, X_{14} + X_{23}\},$$

$$I_- = \text{span}\{X_{12} - X_{34}, X_{13} + X_{24}, X_{14} - X_{23}\}.$$

A direct computation shows that I_+ and I_- are ideals, $[I_+, I_-] = 0$, and

$$\mathfrak{o}_4 = I_+ \oplus I_-.$$

Therefore $\mathfrak{o}_4$ is not simple.

Now assume $n = 3$ or $n \geq 5$, and let $I \neq 0$ be an ideal of $\mathfrak{g}$. It is enough to show that I contains some basis element X_{pq} . Indeed, if $X_{pq} \in I$ and $r \notin \{p, q\}$, then

$$[X_{pq}, X_{qr}] = X_{pr}, \quad [X_{rp}, X_{pq}] = X_{rq},$$

so repeated bracketing gives every basis element of $\mathfrak{g}$.

For $i \neq j$, define

$$P_{ij}(x) = -[X_{ij}, [X_{ij}, x]].$$

Since I is an ideal, $P_{ij}(I) \subseteq I$. Moreover,

$$P_{ij}(X_{kl}) = -[X_{ij}, [X_{ij}, X_{kl}]]$$

and this can be computed directly from the bracket formula.

If $|\{i, j\} \cap \{k, l\}| = 0$, then all Kronecker deltas vanish, so

$$[X_{ij}, X_{kl}] = 0, \quad \text{hence} \quad P_{ij}(X_{kl}) = 0.$$

If $|\{i, j\} \cap \{k, l\}| = 2$, then $\{k, l\} = \{i, j\}$, so $X_{kl} = \pm X_{ij}$. Hence

$$[X_{ij}, X_{kl}] = 0, \quad \text{and again} \quad P_{ij}(X_{kl}) = 0.$$

Now assume $|\{i, j\} \cap \{k, l\}| = 1$. There are four possibilities:

$$\begin{aligned} [X_{ij}, X_{jl}] &= X_{il}, & P_{ij}(X_{jl}) &= -[X_{ij}, X_{il}] = X_{jl}, \\ [X_{ij}, X_{il}] &= -X_{jl}, & P_{ij}(X_{il}) &= -[X_{ij}, -X_{jl}] = X_{il}, \\ [X_{ij}, X_{ki}] &= X_{jk}, & P_{ij}(X_{ki}) &= -[X_{ij}, X_{jk}] = -X_{ik} = X_{ki}, \\ [X_{ij}, X_{kj}] &= -X_{ki}, & P_{ij}(X_{kj}) &= -[X_{ij}, -X_{ki}] = X_{kj}. \end{aligned}$$

Therefore

$$P_{ij}(X_{kl}) = \begin{cases} X_{kl}, & |\{i, j\} \cap \{k, l\}| = 1, \\ 0, & |\{i, j\} \cap \{k, l\}| = 0 \text{ or } 2. \end{cases}$$

If $n = 3$, write

$$x = c_{12}X_{12} + c_{13}X_{13} + c_{23}X_{23} \in I, \quad x \neq 0.$$

Then

$$x - P_{12}(x) = c_{12}X_{12}, \quad x - P_{13}(x) = c_{13}X_{13}, \quad x - P_{23}(x) = c_{23}X_{23}.$$

Since not all c_{ij} vanish, one of these is a non-zero scalar multiple of a basis element. Hence I contains a basis element, so $I = \mathfrak{o}_3$.

Assume now $n \geq 5$. Choose

$$0 \neq x = \sum_{1 \leq a < b \leq n} c_{ab}X_{ab} \in I,$$

and relabel indices so that $c_{12} \neq 0$. Set

$$x_1 = P_{13}(x), \quad x_2 = P_{24}(x_1), \quad x_3 = P_{15}(x_2), \quad x_4 = P_{25}(x_3).$$

By the description of P_{ij} ,

$$x_2 \in \text{span}\{X_{12}, X_{14}, X_{23}, X_{34}\},$$

$$x_3 \in \text{span}\{X_{12}, X_{14}\},$$

and therefore

$$x_4 = c_{12}X_{12} \in I.$$

Thus $X_{12} \in I$, hence $I = \mathfrak{o}_n$.

Therefore $\mathfrak{o}_n$ is simple exactly when $n = 3$ or $n \geq 5$. □

The simplicity of $\mathfrak{sp}_{2n}$

Proof. If $n = 1$, then $\mathfrak{sp}_2(F) \cong \mathfrak{sl}_2(F)$, which is simple. So assume $n \geq 2$, and let

$$\mathfrak{g} = \mathfrak{sp}_{2n}(F) = \left\{ \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} \middle| B^T = B, C^T = C \right\}.$$

For $1 \leq i, j \leq n$, define

$$A_{ij} = E_{ij} - E_{n+j, n+i}, \quad B_{ij} = E_{i, n+j} + E_{j, n+i}, \quad C_{ij} = E_{n+i, j} + E_{n+j, i}.$$

Then $B_{ij} = B_{ji}$ and $C_{ij} = C_{ji}$, and a basis of $\mathfrak{g}$ is

$$\{A_{ij} \mid i \neq j\} \cup \{H_i := A_{ii} \mid 1 \leq i \leq n\} \cup \{B_{ij}, C_{ij} \mid 1 \leq i \leq j \leq n\}.$$

Let $\mathfrak{h} = \text{span}\{H_1, \dots, H_n\}$. A direct computation gives

$$[H_r, A_{ij}] = (\delta_{ri} - \delta_{rj})A_{ij},$$

$$[H_r, B_{ij}] = (\delta_{ri} + \delta_{rj})B_{ij}, \quad [H_r, C_{ij}] = -(\delta_{ri} + \delta_{rj})C_{ij}.$$

Thus every basis element outside $\mathfrak{h}$ is a common eigenvector for $\text{ad } \mathfrak{h}$.

Let $I \neq 0$ be an ideal of $\mathfrak{g}$. Choose $0 \neq x \in I$. If $x \notin \mathfrak{h}$, then exactly as in the proof for $\mathfrak{sl}_n$, a Vandermonde argument with a suitable $h \in \mathfrak{h}$ isolates a non-zero basis vector in I . If $x \in \mathfrak{h}$, write

$$x = \lambda_1 H_1 + \dots + \lambda_n H_n,$$

and choose i with $\lambda_i \neq 0$. Then

$$[x, B_{ii}] = 2\lambda_i B_{ii} \in I.$$

Hence in all cases I contains one of the basis elements A_{ij} ($i \neq j$), B_{ij} , or C_{ij} .

We now show that this already implies $I = \mathfrak{g}$.

First suppose that $B_{ii} \in I$ for some i . Then for every $j \neq i$,

$$A_{ij} = \frac{1}{2}[B_{ii}, C_{ij}] \in I, \quad B_{ij} = \frac{1}{2}[A_{ij}, B_{jj}] \in I, \quad C_{ij} = -\frac{1}{2}[A_{ij}, C_{ii}] \in I.$$

Also,

$$H_i = \frac{1}{4}[B_{ii}, C_{ii}] \in I.$$

For $p, q \neq i$ with $p \neq q$, we then have

$$A_{pq} = [B_{pi}, C_{iq}] \in I, \quad B_{pq} = [A_{pi}, B_{iq}] \in I, \quad C_{pq} = -[A_{ip}, C_{iq}] \in I.$$

For $p \neq i$, the same computations give

$$B_{pp} = [A_{pi}, B_{ip}] \in I, \quad C_{pp} = -[A_{ip}, C_{ip}] \in I.$$

Finally, for $p \neq i$,

$$H_i + H_p = [B_{ip}, C_{ip}] \in I,$$

so $H_p \in I$. Therefore every basis element of $\mathfrak{g}$ lies in I , and hence $I = \mathfrak{g}$.

If $B_{ij} \in I$ with $i \neq j$, then

$$H_i + H_j = [B_{ij}, C_{ij}] \in I,$$

and hence

$$B_{ii} = \frac{1}{2}[H_i + H_j, B_{ii}] \in I.$$

So we again reduce to the first case.

If $A_{ij} \in I$ with $i \neq j$, then

$$B_{ij} = \frac{1}{2}[A_{ij}, B_{jj}] \in I,$$

and so

$$H_i + H_j = [B_{ij}, C_{ij}] \in I.$$

Consequently,

$$B_{ii} = \frac{1}{2}[H_i + H_j, B_{ii}] \in I,$$

and we reduce to the previous case.

If $C_{ii} \in I$, then

$$H_i = -\frac{1}{4}[C_{ii}, B_{ii}] \in I,$$

so

$$B_{ii} = \frac{1}{2}[H_i, B_{ii}] \in I.$$

Again we reduce to the first case.

If $C_{ij} \in I$ with $i \neq j$, then

$$H_i + H_j = [B_{ij}, C_{ij}] \in I,$$

hence

$$C_{ii} = -\frac{1}{2}[H_i + H_j, C_{ii}] \in I.$$

Also,

$$H_i = -\frac{1}{4}[C_{ii}, B_{ii}] \in I,$$

so

$$B_{ii} = \frac{1}{2}[H_i, B_{ii}] \in I.$$

Again we reduce to the first case.

Therefore every non-zero ideal of $\mathfrak{g}$ is equal to $\mathfrak{g}$, so $\mathfrak{sp}_{2n}(F)$ is simple. □

The simplicity of $W(n)$

Proof. Let

$$W(n) = \text{Der}(F[x_1, \dots, x_n]) = \left\{ \sum_{i=1}^n f_i \partial_i \mid f_i \in F[x_1, \dots, x_n] \right\}, \quad \partial_i = \frac{\partial}{\partial x_i},$$

where $\text{char } F = 0$. This Lie algebra is non-abelian, since

$$[\partial_i, x_i] = 1 \neq 0.$$

Let $I \neq 0$ be an ideal of $W(n)$. We prove that $I = W(n)$.

Choose

$$0 \neq D = \sum_{i=1}^n f_i \partial_i \in I.$$

By replacing D with repeated brackets $[\partial_k, D]$, we may successively differentiate the coefficients. Since $\text{char } F = 0$, differentiating a non-constant polynomial lowers its degree without annihilating its highest term. Hence, after finitely many such brackets, we obtain a non-zero constant vector field

$$C = \sum_{i=1}^n c_i \partial_i \in I.$$

Choose k with $c_k \neq 0$. Then for every j ,

$$[x_k \partial_j, C] = \sum_{i=1}^n c_i [x_k \partial_j, \partial_i] = -c_k \partial_j \in I.$$

Therefore $\partial_j \in I$ for all $j = 1, \dots, n$.

Now let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ and fix j . Since $\text{char } F = 0$, the scalar $\alpha_1 + 1$ is invertible in F . Set

$$E = -\frac{1}{\alpha_1 + 1} x_1^{\alpha_1+1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \partial_j \in W(n).$$

Because $\partial_1 \in I$ and I is an ideal,

$$[E, \partial_1] = -\partial_1 \left(-\frac{1}{\alpha_1 + 1} x_1^{\alpha_1+1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \right) \partial_j = x_1^{\alpha_1} \cdots x_n^{\alpha_n} \partial_j \in I.$$

Thus every monomial derivation $x^\alpha \partial_j$ lies in I . These elements form a basis of $W(n)$, so $I = W(n)$.

Therefore $W(n)$ is simple. □

4.2 Homework week 2

Exercise 4.2.1. *Prove that if A is a finite-dimensional associative algebra and for every element $a \in A$ one has $a^n = 0$ for some n , then the algebra A is nilpotent. Do not use Wedderburn's theorems. Use Engel's theorem.*

Proof. Let A be a finite-dimensional associative algebra over a field, such that every element $a \in A$ is nilpotent. We want to show that A is nilpotent as an algebra, meaning there exists some integer $N > 0$ such that $A^N = 0$.

Consider the left regular representation of A on itself. For each $a \in A$, define the left multiplication map $L_a : A \rightarrow A$ by $L_a(x) = ax$ for all $x \in A$. Since A is an associative algebra, the mapping $a \mapsto L_a$ is an algebra homomorphism from A to the algebra of endomorphisms $\text{End}(A)$.

We can view $\text{End}(A)$ as a Lie algebra $\mathfrak{gl}(A)$ with the standard commutator bracket $[T, S] = T \circ S - S \circ T$. The set of all left multiplication maps $\mathfrak{g} = \{L_a \mid a \in A\}$ forms a Lie subalgebra of $\mathfrak{gl}(A)$, because

$$[L_a, L_b] = L_a \circ L_b - L_b \circ L_a = L_{ab} - L_{ba} = L_{[a, b]},$$

and $a, b \in A \implies [a, b] \in A$, so $L_{[a,b]} \in \mathfrak{g}$.

By hypothesis, for every $a \in A$, there exists an integer $n > 0$ such that $a^n = 0$. Consequently, the endomorphism L_a is strictly nilpotent, since $L_a^n(x) = a^n x = 0$ for all $x \in A$. Therefore, $\mathfrak{g}$ is a Lie subalgebra of $\mathfrak{gl}(A)$ consisting entirely of nilpotent endomorphisms.

According to Engel's Theorem, if a Lie subalgebra of $\mathfrak{gl}(V)$ (where V is a non-zero finite-dimensional vector space) consists of nilpotent elements, then there exists a flag of subspaces

$$0 = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_k = A$$

such that $\mathfrak{g}(V_i) \subseteq V_{i-1}$ for all $1 \leq i \leq k$.

Translating this back to the associative algebra A , we have $L_a(V_i) \subseteq V_{i-1}$ for all $a \in A$, which is precisely equivalent to the statement $A \cdot V_i \subseteq V_{i-1}$.

Using this property repeatedly, we obtain:

$$A \cdot V_k \subseteq V_{k-1}$$

$$A^2 \cdot V_k = A \cdot (A \cdot V_k) \subseteq A \cdot V_{k-1} \subseteq V_{k-2}$$

...

$$A^k \cdot V_k \subseteq V_0 = 0.$$

Since $V_k = A$, we conclude that $A^{k+1} = A^k \cdot A \subseteq 0$. Thus, A is a nilpotent associative algebra. $\square$

Exercise 4.2.2. Let $\text{char } F = 0$; let L be a finite-dimensional solvable Lie algebra; let $e_1, \dots, e_m$ be a basis of the algebra L . Let M be a finite-dimensional L -module. Suppose that every element e_i , for $1 \leq i \leq m$, acts on M nilpotently. Prove that the algebra L acts on M nilpotently.

Proof. Let $\bar{F}$ be the algebraic closure of F . We consider the scalar extensions of the Lie algebra and the module: $\bar{L} = L \otimes_F \bar{F}$ and $\bar{M} = M \otimes_F \bar{F}$.

Since L is a finite-dimensional solvable Lie algebra over F , its scalar extension $\bar{L}$ remains a solvable Lie algebra over $\bar{F}$. Furthermore, the elements $e_1, \dots, e_m$ naturally form a basis for $\bar{L}$ as a vector space over $\bar{F}$.

Because $\text{char } \bar{F} = 0$ and $\bar{F}$ is algebraically closed, we can apply Lie's Theorem to the finite-dimensional $\bar{L}$ -module $\bar{M}$. Lie's Theorem guarantees the existence of a basis for $\bar{M}$ such that the representation $\rho : \bar{L} \rightarrow \mathfrak{gl}(\bar{M})$ maps every element of $\bar{L}$ to an upper triangular matrix.

Consequently, for each $x \in \bar{L}$, the diagonal entries of the matrix $\rho(x)$ are given by a set of linear functionals $\lambda_1(x), \dots, \lambda_n(x)$, where $n = \dim_{\bar{F}} \bar{M}$ and $\lambda_k \in \bar{L}^*$.

We are given that each basis element e_i ($1 \leq i \leq m$) acts nilpotently on M . This property is preserved under scalar extension, so each e_i also acts nilpotently on $\bar{M}$. An upper triangular matrix that is nilpotent must have all its diagonal entries equal to zero. Thus, for each $i \in \{1, \dots, m\}$ and each $k \in \{1, \dots, n\}$, we have:

$$\lambda_k(e_i) = 0.$$

Since the linear functionals λ_k vanish on the basis $e_1, \dots, e_m$, and linear transformations are determined by their action on a basis, they must vanish identically on the entire Lie algebra $\bar{L}$. That is, $\lambda_k(x) = 0$ for all $x \in \bar{L}$.

This implies that for any $x \in \bar{L}$, the matrix $\rho(x)$ is strictly upper triangular. Consequently, every $x \in \bar{L}$ acts as a nilpotent endomorphism on $\bar{M}$.

Restricting this result back to the original base field F and the original module M , we conclude that every element $x \in L$ acts nilpotently on M . By Engel's Theorem, a Lie algebra of nilpotent endomorphisms acts nilpotently, which completes the proof. $\square$

Exercise 4.2.3. For every prime p , find an example showing that Lie's theorem fails.

Proof. Let F be an algebraically closed field of characteristic $p > 0$. We will construct a finite-dimensional solvable Lie algebra $\mathfrak{g}$ and a finite-dimensional representation V over F such that $\mathfrak{g}$ has no common eigenvector in V , thus showing that Lie's theorem fails.

Let V be a p -dimensional vector space over F with a chosen basis $v_0, v_1, \dots, v_{p-1}$. We define two linear transformations $x, y \in \mathfrak{gl}(V)$ by defining their actions on the basis vectors:

$$\begin{aligned} x(v_i) &= v_{i-1} \quad \text{for } 0 \leq i \leq p-1, \\ y(v_i) &= iv_i \quad \text{for } 0 \leq i \leq p-1, \end{aligned}$$

where the subscripts for the action of x are taken modulo p (so $x(v_0) = v_{p-1}$).

Let $\mathfrak{g}$ be the Lie subalgebra of $\mathfrak{gl}(V)$ generated by x and y . First, we compute the commutator $[x, y]$ acting on an arbitrary basis vector v_i :

$$[x, y](v_i) = x(y(v_i)) - y(x(v_i)) = x(iv_i) - y(v_{i-1}) = iv_{i-1} - (i-1)v_{i-1} = v_{i-1} = x(v_i).$$

Since this holds for all basis vectors, we have $[x, y] = x$. This shows that the vector space spanned by x and y is closed under the Lie bracket, so $\mathfrak{g} = \text{span}_F\{x, y\}$ is a 2-dimensional Lie algebra. Furthermore, its derived algebra is $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}] = \text{span}_F\{x\}$. Since $\mathfrak{g}'$ is 1-dimensional, it is abelian, which implies that $\mathfrak{g}$ is a solvable Lie algebra.

Next, we show that $\mathfrak{g}$ has no common eigenvector in V . Suppose, for the sake of contradiction, that $v = \sum_{i=0}^{p-1} c_i v_i$ is a common non-zero eigenvector for both x and y . Since v is an eigenvector of y , and the eigenvalues of y are exactly $0, 1, \dots, p-1$ (which are all distinct in F because the characteristic is p), v must belong to one of the 1-dimensional eigenspaces of y . Thus, v must be a non-zero scalar multiple of v_k for some $0 \leq k \leq p-1$.

However, v must also be an eigenvector for x . We check the action of x on v_k :

$$x(v_k) = v_{k-1} \pmod{p}.$$

Because p is a prime (and thus $p \geq 2$), v_{k-1} is linearly independent from v_k . Therefore, v_k is not an eigenvector for x , which is a contradiction.

Consequently, the solvable Lie algebra $\mathfrak{g}$ has no common eigenvector in the representation V , demonstrating that Lie's theorem does not hold in characteristic p . $\square$